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IPRSOCP: A Primal-Dual Interior-Point Relaxation Algorithm for Second-Order Cone Programming

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Abstract

Inspired by the smoothing barrier augmented Lagrangian function in Liu et al. (Math Methods Oper Res 96(3):351–382, 2022), we propose a primal-dual interior-point relaxation algorithm for second-order cone programming, called IPRSOCP. Two features of the IPRSOCP algorithm are as follows. One is that the iterative points of the proposed algorithm need not lie inside the interior region, convening the use of warm-start. The other is that an explicit form of the Schur complement matrix is explored such that the low-rank structure of the Schur complement matrix can be used to improve numerical stability and efficiency. Under suitable assumptions, it is shown that the barrier parameter in the IPRSOCP algorithm tends to zero and the generated sequence of iterations is globally convergent to the solution. Numerical results demonstrate

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that the IPRSOCP algorithm is competitive with the open-source benchmark solvers, SeDuMi, SDPT3, and ECOS, in terms of robustness and efficiency.

Keywords Second-order cone programming · Interior-point relaxation method · Smoothing barrier augmented Lagrangian function · Schur complement matrix · Global convergence

Mathematics Subject Classification 90C25 · 90C51 · 65K05

1 Introduction

Second-order cone programming (SOCP) has received a wide range of applications. It can be used to model a variety of optimization problems, including robust optimization [3, 14], portfolio optimization [4], trajectory optimization [26], and others [8, 31, 35]. As SOCP is a class of fundamental optimization problems, there have been many efficient interior-point and non-interior-point methods among many others for the solution of SOCP.

Interior-point methods are efficient to solve SOCP problems (see [2, 24, 29, 36, 41, 44, 47]). The approximate optimal solution can be obtained by iteratively solving a sequence of barrier problems with gradually decreasing barrier parameters. One of the advantages of interior-point methods is the polynomial iteration complexity [37, 46], which enables for highly accurate solutions with a few number of iterations. Many state-of-the-art open-source solvers, such as SeDuMi [43], SDPT3 [45], and ECOS [13], are developed based on interior-point methods. Some application problems, including multi-phase trajectory optimization [26], require solving a series of similar SOCP problems. Warm-start can utilize information from previous subproblems, reducing the number of iterations and improving overall efficiency. Performing warm-start for interior-point methods is difficult as the iterative points must be inside the interior region, despite the numerous studies on this topic [21, 22, 25, 49]. Consequently, non-interior-point methods, which do not require iterative points inside the interior region, have gained considerable interest.

Non-interior-point methods, also known as smoothing Newton methods, have been applied to solve SOCP problems. These methods can be traced back to the semismooth Newton method for solving nonsmooth equations (see [11, 16, 28, 40] and the references therein). Introducing smoothing functions can further enhance the performance of the semismooth Newton method. Commonly used smoothing functions include the Chen–Harker–Kanzow–Smale (CHKS) function [6, 27, 42] and the Fischer–Burmeister (FB) function suggested by Kanzow [27]. Fukushima et al. [17] investigated the theoretical properties of some smoothing functions using Jordan algebra. In addition, regularization methods have been integrated into the smoothing Newton method to address ill-posed problems [23, 30]. Other research on non-interior-point methods for SOCP problems can be found in [7, 9, 38].

The most time-consuming parts of both interior-point methods and non-interior-point methods are the formulation and factorization of Schur complement matrix. Some studies utilized the explicit form of Schur complement matrix to make the factorization more efficient and stable. One such method is the product-form Cholesky factorization (PFCF) approach presented by Goldfarb and Scheinberg [19]. Furthermore, the expanded sparse representation of the Schur complement matrix defined by Domahidi et al. [13] increases the size of the matrix while greatly reducing its density. Another method is the reduced augmented equation approach introduced by Cai and Toh [5], which addresses numerical challenges associated with the ill-posed Schur complement matrices. These are some effective ways of dealing with Schur complement matrices in interior-point methods. To the best of our knowledge, no research is available on the explicit form of Schur complement matrices for non-interior-point methods.

Some studies have combined the interior-point technique with the Hestenes–Powell augmented Lagrangian function to solve optimization problems. For example, Goldfarb and Scheinberg [20] introduced the modified barrier augmented Lagrangian (MBAL) method, which features Q-linear or even Q-superlinear convergence. The MBAL method employs a modified barrier function rather than the barrier function directly. Robinson [18] developed a primal-dual shifted penalty-barrier method that introduces additional variables. The primal-dual interior-point relaxation methods (IPRM) [10, 32–34, 51] were proposed based on the smoothing barrier augmented Lagrangian (SBAL) function. The design of our algorithm is motivated by the good theoretical properties and practical performance of IPRM. The optimality conditions derived from the SBAL and CHKS functions share a similar structure, with the exception that the former incorporates an additional penalty parameter. Recently, IPRM has also been successfully applied to the positive semidefinite cone [50]. The SOCP problem can be solved as a semidefinite programming problem. However, it is more efficient to solve the SOCP problem directly. As a result, we design a special algorithm, called IPRSOCP, for SOCP problems.

The contributions in this paper can be summarized as follows:

- We extend the smoothing barrier augmented Lagrangian function in IPRM to second-order cones. Using Jordan algebra, we analyze the differential and Lipschitzian properties of the functions involved. IPRSOCP is derived from this smoothing barrier augmented Lagrangian function, while most existing non-interior-point methods are based on CHKS or FB functions.
- We provide an explicit form of the Schur complement matrix for IPRSOCP that existing non-interior-point methods do not. This explicit form contains a low-rank structure, which is utilized to improve the efficiency of matrix factorization. In particular, the implicit barrier parameter in the Schur complement matrix can be extracted, thereby improving the condition number of the Schur complement matrix and enhancing the numerical stability of the algorithm.
- We perform numerical experiments to compare our algorithm with three state-of-the-art open-source solvers: SeDuMi, SDPT3, and ECOS. The numerical results demonstrate that our algorithm is comparable to them in terms of robustness and efficiency.

The rest of this paper is organized as follows: Sect. 2 introduces some preliminary knowledge about second-order cones. In Sect. 3, we construct a smoothing barrier augmented Lagrangian function and provide the differential and Lipschitzian properties. Section 4 designs the primal-dual interior-point relaxation algorithm and gives the explicit form of Schur complement matrix in this algorithm. In Sect. 5, we establish global convergence under suitable assumptions. The numerical results in Sect. 6 show that IPRSOCP is robust and efficient. Some conclusions are drawn in the last section.

Notation A letter with superscript (k) is related to the k-th iteration. The subscript i indicates the i-th component of a vector. We may omit the superscript or subscript whenever it is obvious. I is an identity matrix whose order is either marked in the subscript or is clear in the context. $\|\boldsymbol{v}\|$ denotes the Euclidean norm of a vector $\boldsymbol{v}$, the ∞-norm of $\boldsymbol{v}$ is denoted as $\|\boldsymbol{v}\|_\infty$. $\mathrm{diag}(\boldsymbol{x})$ is a diagonal matrix with the components of the vector $\boldsymbol{x}$ on the diagonal. $c = \Theta(\mu)$ means that there are positive constants a, b such that $a\mu \leqslant c \leqslant b\mu$ as $\mu \to 0$. We follow the notations in MATLAB, using "," to join two vectors or matrices by rows, and ";" to join two vectors or matrices by columns.

2 Preliminaries

In this section, we present some basic conclusions about second-order cones, which can also be found in [1, 17].

Consider the following second-order cone programming problem:

$$\begin{aligned} \min\ & \sum_{i=1}^{n} \boldsymbol{c}_i^\top \boldsymbol{x}_i \\ \text{s.t.}\ & \sum_{i=1}^{n} A_i \boldsymbol{x}_i = \boldsymbol{b}, \\ & \boldsymbol{x}_i \in K_i,\ i = 1, \cdots, n, \end{aligned} \tag{1}$$

where $A_i \in \mathbb{R}^{m\times k_i}$, $\boldsymbol{c}_i \in \mathbb{R}^{k_i}$, $\boldsymbol{b} \in \mathbb{R}^m$, K_i is a second-order cone of dimension k_i, that is,

$$K_i = \{\boldsymbol{x}_i = (x_{i0}; \bar{\boldsymbol{x}}_i) \in \mathbb{R} \times \mathbb{R}^{k_i - 1} : \|\bar{\boldsymbol{x}}_i\| \leqslant x_{i0}\}. \tag{2}$$

The interior of the i-th second-order cone is defined by

$$\mathrm{int}(K_i) = \{\boldsymbol{x}_i = (x_{i0}; \bar{\boldsymbol{x}}_i) \in \mathbb{R} \times \mathbb{R}^{k_i - 1} : \|\bar{\boldsymbol{x}}_i\| < x_{i0}\}. \tag{3}$$

Furthermore, we denote

$$\begin{aligned} &k = k_1 + k_2 + \cdots + k_n, && K = K_1 \times K_2 \times \cdots \times K_n, \\ &A = (A_1, A_2, \cdots, A_n) \in \mathbb{R}^{m\times k}, && \boldsymbol{c} = (\boldsymbol{c}_1; \boldsymbol{c}_2; \cdots; \boldsymbol{c}_n) \in \mathbb{R}^k, \\ &\boldsymbol{x} = (\boldsymbol{x}_1; \boldsymbol{x}_2; \cdots; \boldsymbol{x}_n) \in K, && \boldsymbol{s} = (\boldsymbol{s}_1; \boldsymbol{s}_2; \cdots; \boldsymbol{s}_n) \in K. \end{aligned}$$

Then problem (1) and its dual can be written as

$$(P)\quad \begin{aligned} &\min\ \boldsymbol{c}^\top \boldsymbol{x} \\ &\text{s.t. } A\boldsymbol{x} = \boldsymbol{b}, \\ &\quad\ \boldsymbol{x} \in K, \end{aligned} \qquad (D)\quad \begin{aligned} &\max\ \boldsymbol{b}^\top \boldsymbol{\lambda} \\ &\text{s.t. } A^\top \boldsymbol{\lambda} + \boldsymbol{s} = \boldsymbol{c}, \\ &\quad\ \boldsymbol{s} \in K. \end{aligned} \tag{4}$$

Consider the case where there is a single cone, i.e., $n = 1$. For any $\boldsymbol{x} = (x_0; \bar{\boldsymbol{x}}) \in \mathbb{R} \times \mathbb{R}^{k-1}$, the arrow-shaped matrix $\mathrm{Arw}(\boldsymbol{x})$ is denoted by

$$\mathrm{Arw}(\boldsymbol{x}) := \begin{pmatrix} x_0 & \bar{\boldsymbol{x}}^\top \\ \bar{\boldsymbol{x}} & x_0 I_{k-1} \end{pmatrix}.$$

Denote the Jordan product of two vectors $\boldsymbol{x} = (x_0; \bar{\boldsymbol{x}}) \in \mathbb{R} \times \mathbb{R}^{k-1}$ and $\boldsymbol{s} = (s_0; \bar{\boldsymbol{s}}) \in \mathbb{R} \times \mathbb{R}^{k-1}$ by using the arrow-shaped matrix as

$$\boldsymbol{x} \circ \boldsymbol{s} := \begin{pmatrix} \boldsymbol{x}^\top \boldsymbol{s} \\ x_0 \bar{\boldsymbol{s}} + s_0 \bar{\boldsymbol{x}} \end{pmatrix} = \mathrm{Arw}(\boldsymbol{x})\boldsymbol{s} = \mathrm{Arw}(\boldsymbol{x})\mathrm{Arw}(\boldsymbol{s})\boldsymbol{e},$$

where $\boldsymbol{e} = (1; 0; \cdots; 0) \in \mathbb{R}^k$ is the unit element of the k-dimension second-order cone.

The spectral decomposition of a vector $\boldsymbol{x}$ can be defined by

$$\boldsymbol{x} = \eta_1 \boldsymbol{d}_1 + \eta_2 \boldsymbol{d}_2, \tag{5}$$

where η_1 and η_2 are the spectral values. The associated spectral vectors $\boldsymbol{d}_1$ and $\boldsymbol{d}_2$ are given by

$$\begin{aligned} &\eta_1 = x_0 + \|\bar{\boldsymbol{x}}\|, && \eta_2 = x_0 - \|\bar{\boldsymbol{x}}\|, \\ &\boldsymbol{d_1} = \begin{cases} \frac{1}{2}\left(1; \frac{\bar{\boldsymbol{x}}}{\|\bar{\boldsymbol{x}}\|}\right), & \text{if } \bar{\boldsymbol{x}} \neq 0, \\ \frac{1}{2}(1; \boldsymbol{\omega}), & \text{if } \bar{\boldsymbol{x}} = 0, \end{cases} && \boldsymbol{d_2} = \begin{cases} \frac{1}{2}\left(1; -\frac{\bar{\boldsymbol{x}}}{\|\bar{\boldsymbol{x}}\|}\right), & \text{if } \bar{\boldsymbol{x}} \neq 0, \\ \frac{1}{2}(1; -\boldsymbol{\omega}), & \text{if } \bar{\boldsymbol{x}} = 0, \end{cases} \end{aligned}$$

$\boldsymbol{\omega}$ is any vector in $\mathbb{R}^{k-1}$ satisfying $\|\boldsymbol{\omega}\| = 1$.

It is easy to verify that the spectral vectors of $\boldsymbol{x}$ have the following properties:

$$\boldsymbol{d}_1 \circ \boldsymbol{d}_2 = 0,\ \boldsymbol{d}_1 \circ \boldsymbol{d}_1 = \boldsymbol{d}_1,\ \boldsymbol{d}_2 \circ \boldsymbol{d}_2 = \boldsymbol{d}_2,\ \boldsymbol{d}_1 + \boldsymbol{d}_2 = \boldsymbol{e}. \tag{6}$$

Any pair of vectors $\{\boldsymbol{d}_1, \boldsymbol{d}_2\}$ that satisfies (6) is called a *Jordan frame*. Obviously, there is $\boldsymbol{x} \circ \boldsymbol{s} = \boldsymbol{s} \circ \boldsymbol{x}$ for any $\boldsymbol{x}, \boldsymbol{s} \in \mathbb{R}^n$, but $\mathrm{Arw}(\boldsymbol{x})\mathrm{Arw}(\boldsymbol{s}) = \mathrm{Arw}(\boldsymbol{s})\mathrm{Arw}(\boldsymbol{x})$ is generally not satisfied. To derive the commutative property of the arrow matrices, we introduce the concept of operator commutativity. We say that $\boldsymbol{x}$ and $\boldsymbol{s}$ are *operator commutative* if they share a Jordan frame, that is,

$$\boldsymbol{x} = \eta_1 \boldsymbol{d}_1 + \eta_2 \boldsymbol{d}_2,\ \boldsymbol{s} = \beta_1 \boldsymbol{d}_1 + \beta_2 \boldsymbol{d}_2$$

for a Jordan frame $\{\boldsymbol{d}_1, \boldsymbol{d}_2\}$.

The following conclusion about operator commutativity can also be found in [1].

Lemma 1 *The following statements are equivalent*:

1. $\boldsymbol{x}$ *and* $\boldsymbol{s}$ *are operator commutative*;
2. *Arw*$(\boldsymbol{x})$ *and Arw*$(\boldsymbol{s})$ *are commutative.*

It is obvious that $\boldsymbol{x} \in K$ if and only if $\eta_1, \eta_2 \geqslant 0$, and $\boldsymbol{x} \in \text{int}(K)$ if and only if $\eta_1, \eta_2 > 0$. The eigenvalues of $\text{Arw}(\boldsymbol{x})$ have the following properties:

Theorem 1 $\eta_1 = x_0 - \|\bar{\boldsymbol{x}}\|$ *and* $\eta_2 = x_0 + \|\bar{\boldsymbol{x}}\|$ *are eigenvalues of Arw*$(\boldsymbol{x})$. *If* $\eta_1 \neq \eta_2$, *then each one has multiplicity one*; *the corresponding eigenvectors are* $\boldsymbol{d}_1$ *and* $\boldsymbol{d}_2$, *respectively. Furthermore,* x_0 *is an eigenvalue of Arw*$(\boldsymbol{x})$ *and has a multiplicity of* $k-2$ *when* $\boldsymbol{x} \neq \boldsymbol{0}$.

Based on the spectral decomposition of $\boldsymbol{x}$, the trace and determinant of $\boldsymbol{x}$ can be defined by

$$\begin{aligned} \text{trace}(\boldsymbol{x}) &= \eta_1 + \eta_2 = 2x_0, \\ \det(\boldsymbol{x}) &= \eta_1\eta_2 = x_0^2 - \|\bar{\boldsymbol{x}}\|^2. \end{aligned} \tag{7}$$

It is easy to verify

$$\begin{aligned} \boldsymbol{x}^2 := \boldsymbol{x} \circ \boldsymbol{x} &= (\eta_1\boldsymbol{d}_1 + \eta_2\boldsymbol{d}_2) \circ (\eta_1\boldsymbol{d}_1 + \eta_2\boldsymbol{d}_2) \\ &= \eta_1^2\boldsymbol{d}_1 \circ \boldsymbol{d}_1 + \eta_1\eta_2\boldsymbol{d}_1 \circ \boldsymbol{d}_2 + \eta_2\eta_1\boldsymbol{d}_2 \circ \boldsymbol{d}_1 + \eta_2^2\boldsymbol{d}_2 \circ \boldsymbol{d}_2 \\ &= \eta_1^2\boldsymbol{d}_1 + \eta_2^2\boldsymbol{d}_2. \end{aligned} \tag{8}$$

Furthermore, we define $\boldsymbol{x}^0 := \boldsymbol{e}$.

If $\boldsymbol{x} \in K$, then denote the square root of $\boldsymbol{x}$,

$$\boldsymbol{x}^{\frac{1}{2}} := \eta_1^{\frac{1}{2}}\boldsymbol{d}_1 + \eta_2^{\frac{1}{2}}\boldsymbol{d}_2, \tag{9}$$

and the inverse of $\boldsymbol{x}$ if $\boldsymbol{x} \in \text{int}(K)$

$$\boldsymbol{x}^{-1} := \eta_1^{-1}\boldsymbol{d}_1 + \eta_2^{-1}\boldsymbol{d}_2 = \frac{1}{\det(\boldsymbol{x})}R\boldsymbol{x}, \text{ where } R = \begin{pmatrix} 1 & \boldsymbol{0}^\top \\ \boldsymbol{0} & -I_{k-1} \end{pmatrix}. \tag{10}$$

The above definitions can be extended to multiple cones. For two matrices A and B, we define the notation $A \oplus B$ as

$$A \oplus B = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

Suppose that $\boldsymbol{x} = (\boldsymbol{x}_1; \cdots; \boldsymbol{x}_n)$, $\boldsymbol{s} = (\boldsymbol{s}_1; \cdots; \boldsymbol{s}_n)$, and $\boldsymbol{x}_i, \boldsymbol{s}_i \in \mathbb{R}^{k_i}$, $\boldsymbol{e}_i = (1, \boldsymbol{0}) \in \mathbb{R}^{k_i}$ is the unit element of the i-th second-order cone for $i = 1, \cdots, n$. Then we define

$$\boldsymbol{x} \circ \boldsymbol{s} := (\boldsymbol{x}_1 \circ \boldsymbol{s}_1; \cdots; \boldsymbol{x}_n \circ \boldsymbol{s}_n); \tag{11a}$$
$$\operatorname{Arw}(\boldsymbol{x}) := \operatorname{Arw}(\boldsymbol{x}_1) \oplus \operatorname{Arw}(\boldsymbol{x}_2) \oplus \cdots \oplus \operatorname{Arw}(\boldsymbol{x}_n); \tag{11b}$$
$$\boldsymbol{x}^{-1} := \left(\boldsymbol{x}_1^{-1}; \cdots; \boldsymbol{x}_n^{-1}\right); \tag{11c}$$
$$\boldsymbol{e} := (\boldsymbol{e}_1; \cdots; \boldsymbol{e}_n). \tag{11d}$$

3 A Smoothing Barrier Augmented Lagrangian

In this section, we introduce a smoothing barrier augmented Lagrangian (SBAL) function by combining the logarithmic barrier function with the Hestenes–Powell augmented Lagrangian function. Following this, we provide the differential and Lipschitzian properties of relevant functions.

By introducing a new variable $\boldsymbol{z} = (z_1; \cdots; z_n) \in K$, $z_i = (z_{i0}; \bar{z}_i) \in \mathbb{R} \times \mathbb{R}^{k_i-1}$, $i = 1, \cdots, n$, an equivalent problem of (1) can be obtained

$$\begin{aligned} \min\ & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.}\ & A\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} = \boldsymbol{z},\ \boldsymbol{z} \in K. \end{aligned} \tag{12}$$

Like interior-point methods, a subproblem with the logarithmic barrier function is solved at each iteration:

$$\begin{aligned} \min\ & \boldsymbol{c}^\top \boldsymbol{x} - \frac{\mu}{2} \sum_{i=1}^{n} \ln \det(z_i) \\ \text{s.t.}\ & A\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} = \boldsymbol{z}, \end{aligned} \tag{13}$$

where $\mu > 0$ is the barrier parameter. Then, we apply the augmented Lagrangian function to handle the newly introduced constraint $\boldsymbol{x} = \boldsymbol{z}$,

$$\begin{aligned} \min\ & \boldsymbol{c}^\top \boldsymbol{x} - \frac{\mu}{2} \sum_{i=1}^{n} \ln \det(z_i) - \boldsymbol{s}^\top (\boldsymbol{x} - \boldsymbol{z}) + \frac{\rho}{2} \sum_{i=1}^{n} \|\boldsymbol{x}_i - z_i\|_2^2 \\ \text{s.t.}\ & A\boldsymbol{x} = \boldsymbol{b}, \end{aligned} \tag{14}$$

where $\boldsymbol{s} \in K$ is the Lagrange multiplier and $\rho > 0$ is the penalty parameter.

We define the objective function of problem (14) as $F(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{s}; \mu, \rho)$. This function combines the barrier function with the augmented Lagrangian function. The barrier parameter corresponds to the smoothing parameter used in the smoothed CHKS function [40]. Hence, $F(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{s}; \mu, \rho)$ is referred to as the smoothing barrier augmented Lagrangian function.

We can get the minimizer of variable z_i by taking its derivative for $i = 1, \cdots, n$,

$$\nabla_{z_i} F(\boldsymbol{x}, z, \boldsymbol{s}; \mu, \rho) = -\mu z_i^{-1} + \boldsymbol{s}_i + \rho(z_i - \boldsymbol{x}_i) = \boldsymbol{0}. \tag{15}$$

Equation (15) has the following explicit solution inside the second-order cone:

$$z_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho) = \frac{\left((\boldsymbol{s}_i - \rho \boldsymbol{x}_i)^2 + 4\rho\mu \boldsymbol{e}_i\right)^{\frac{1}{2}} - (\boldsymbol{s}_i - \rho \boldsymbol{x}_i)}{2\rho}. \tag{16}$$

In addition, we define $\boldsymbol{y}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho)$, $\boldsymbol{v}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho)$ and $\boldsymbol{u}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho)$ for brevity, which have the following form:

$$\boldsymbol{y}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho) = \frac{\left((\boldsymbol{s}_i - \rho \boldsymbol{x}_i)^2 + 4\rho\mu \boldsymbol{e}_i\right)^{\frac{1}{2}} + (\boldsymbol{s}_i - \rho \boldsymbol{x}_i)}{2\rho}, \tag{17a}$$

$$\boldsymbol{v}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho) = \boldsymbol{s}_i - \rho \boldsymbol{x}_i, \tag{17b}$$

$$\boldsymbol{u}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho) = \left(\boldsymbol{v}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho)^2 + 4\rho\mu \boldsymbol{e}_i\right)^{\frac{1}{2}}. \tag{17c}$$

Without ambiguity, the four functions in (16) and (17) are also abbreviated as z_i, $\boldsymbol{y}_i, \boldsymbol{v}_i, \boldsymbol{u}_i$. Therefore, we have $z_i = \dfrac{\boldsymbol{u}_i - \boldsymbol{v}_i}{2\rho}$ and $\boldsymbol{y}_i = \dfrac{\boldsymbol{u}_i + \boldsymbol{v}_i}{2\rho}$. In addition, we denote $\boldsymbol{y} = (\boldsymbol{y}_1; \boldsymbol{y}_2; \cdots; \boldsymbol{y}_n)$, $\boldsymbol{v} = (\boldsymbol{v}_1; \boldsymbol{v}_2; \cdots; \boldsymbol{v}_n)$, and $\boldsymbol{u} = (\boldsymbol{u}_1; \boldsymbol{u}_2; \cdots; \boldsymbol{u}_n)$.

Lemma 2 and Theorem 2 present the commutative and Lipschitzian properties on z and $\boldsymbol{y}$.

Lemma 2 *Given $\mu \geqslant 0$ and $\rho > 0$, for $i = 1, \cdots, n$, z_i and $\boldsymbol{y}_i$ are defined by (16) and (17). Then the following conclusions hold*:

1. *z_i and $\boldsymbol{y}_i$ are operator commutative, namely*

$$Arw\,(z_i)\,Arw\,(\boldsymbol{y}_i) = Arw\,(\boldsymbol{y}_i)\,Arw\,(z_i)\,,\ i = 1, \cdots, n.$$

2. *$z_i,\ \boldsymbol{y}_i \in K_i$, and $z_i \circ \boldsymbol{y}_i = \dfrac{\mu}{\rho}\boldsymbol{e}_i,\ i = 1, \cdots, n.$*

Proof The proof can be found in Appendix A.

Theorem 2 *Given $\mu_1,\ \mu_2 > 0$, and $\boldsymbol{x}_i = (x_{i0}; \bar{\boldsymbol{x}}_i)$, $\boldsymbol{s}_i = (s_{i0}; \bar{\boldsymbol{s}}_i) \in \mathbb{R} \times \mathbb{R}^{k_i - 1}$ for $i = 1, \cdots, n$, then*

$$\|z_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_1, \rho) - z_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_2, \rho)\| \leqslant \sqrt{\frac{2}{\rho}}|\sqrt{\mu_1} - \sqrt{\mu_2}|, \tag{18a}$$

$$\|\boldsymbol{y}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_1, \rho) - \boldsymbol{y}_i\,(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_2, \rho)\| \leqslant \sqrt{\frac{2}{\rho}}|\sqrt{\mu_1} - \sqrt{\mu_2}|. \tag{18b}$$

Furthermore, if

$$\eta_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i\right)=\min\left\{\left|v_{i0}+\|\bar{\boldsymbol{v}}_i\|\right|,\left|v_{i0}-\|\bar{\boldsymbol{v}}_i\|\right|\right\}>0,$$

then

$$\left\|\boldsymbol{z}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_1, \rho\right)-\boldsymbol{z}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_2, \rho\right)\right\| \leqslant \frac{\sqrt{2}}{\eta_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i\right)}\left|\mu_1-\mu_2\right|, \tag{19a}$$

$$\left\|\boldsymbol{y}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_1, \rho\right)-\boldsymbol{y}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_2, \rho\right)\right\| \leqslant \frac{\sqrt{2}}{\eta_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i\right)}\left|\mu_1-\mu_2\right|. \tag{19b}$$

Proof Given the spectral decomposition $\boldsymbol{v}_i=\eta_1 \boldsymbol{d}_1+\eta_2 \boldsymbol{d}_2$ for a fixed i, then we have

$$\begin{aligned}
&\left\|\boldsymbol{z}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_1, \rho\right)-\boldsymbol{z}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_2, \rho\right)\right\| \\
&=\left\|\frac{\sqrt{\eta_1^2+4\rho\mu_1}-\sqrt{\eta_1^2+4\rho\mu_2}}{2\rho}\boldsymbol{d}_1+\frac{\sqrt{\eta_2^2+4\rho\mu_1}-\sqrt{\eta_2^2+4\rho\mu_2}}{2\rho}\boldsymbol{d}_2\right\| \\
&=\left\|\frac{2\left(\mu_1-\mu_2\right)}{\sqrt{\eta_1^2+4\rho\mu_1}+\sqrt{\eta_1^2+4\rho\mu_2}}\boldsymbol{d}_1+\frac{2\left(\mu_1-\mu_2\right)}{\sqrt{\eta_2^2+4\rho\mu_1}+\sqrt{\eta_2^2+4\rho\mu_2}}\boldsymbol{d}_2\right\| \\
&\leqslant\left|\frac{\sqrt{2}\left(\mu_1-\mu_2\right)}{\sqrt{\eta_1^2+4\rho\mu_1}+\sqrt{\eta_1^2+4\rho\mu_2}}\right|+\left|\frac{\sqrt{2}\left(\mu_1-\mu_2\right)}{\sqrt{\eta_2^2+4\rho\mu_1}+\sqrt{\eta_2^2+4\rho\mu_2}}\right| \\
&\leqslant 2\left|\frac{\mu_1-\mu_2}{\sqrt{2\rho}\left(\sqrt{\mu_1}+\sqrt{\mu_2}\right)}\right| \\
&\leqslant \sqrt{\frac{2}{\rho}}\left|\sqrt{\mu_1}-\sqrt{\mu_2}\right|.
\end{aligned} \tag{20}$$

By the same way, we can obtain

$$\left\|\boldsymbol{y}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_1, \rho\right)-\boldsymbol{y}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_2, \rho\right)\right\| \leqslant \sqrt{\frac{2}{\rho}}\left|\sqrt{\mu_1}-\sqrt{\mu_2}\right|.$$

Suppose $\eta_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i\right)>0$, which together with the first inequality of (20) gives

$$\begin{aligned}
&\left\|\boldsymbol{z}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_1, \rho\right)-\boldsymbol{z}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_2, \rho\right)\right\| \\
&\quad\leqslant\left|\frac{\mu_1-\mu_2}{\sqrt{2}\eta_1}\right|+\left|\frac{\mu_1-\mu_2}{\sqrt{2}\eta_2}\right| \\
&\quad\leqslant \frac{\sqrt{2}}{\eta_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i\right)}\left|\mu_1-\mu_2\right|.
\end{aligned} \tag{21}$$

Similarly, we have

$$\left\|\boldsymbol{y}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_1, \rho\right)-\boldsymbol{y}_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu_2, \rho\right)\right\| \leqslant \frac{\sqrt{2}}{\eta_i\left(\boldsymbol{x}_i, \boldsymbol{s}_i\right)}\left|\mu_1-\mu_2\right|. \tag{22}$$

The proof is completed.

We now present the derivatives of z and y, which will be used in the Newton's method to calculate the search direction.

Lemma 3 *If $\boldsymbol{u}_i \in int(K_i),\ i = 1, \cdots, n$, then we have*

$$\begin{aligned} &\nabla_{\boldsymbol{x}_i} \boldsymbol{z}_i = \rho Arw^{-1}(\boldsymbol{u}_i) Arw(\boldsymbol{z}_i), \ \nabla_{\boldsymbol{x}_i} \boldsymbol{y}_i = -\rho Arw^{-1}(\boldsymbol{u}_i) Arw(\boldsymbol{y}_i), \\ &\nabla_{\boldsymbol{s}_i} \boldsymbol{z}_i = -Arw^{-1}(\boldsymbol{u}_i) Arw(\boldsymbol{z}_i), \ \nabla_{\boldsymbol{s}_i} \boldsymbol{y}_i = Arw^{-1}(\boldsymbol{u}_i) Arw(\boldsymbol{y}_i), \\ &\frac{\partial \boldsymbol{z}_i}{\partial \mu} = Arw^{-1}(\boldsymbol{u}_i)\boldsymbol{e}_i, \qquad \frac{\partial \boldsymbol{y}_i}{\partial \mu} = Arw^{-1}(\boldsymbol{u}_i)\boldsymbol{e}_i. \end{aligned} \tag{23}$$

Proof Note that $\boldsymbol{u}_i^2 = \boldsymbol{v}_i^2 + 4\rho\mu\boldsymbol{e}_i$. Thus, one has

$$2\mathrm{Arw}(\boldsymbol{u}_i)\nabla_{\boldsymbol{x}_i}\boldsymbol{u}_i = -2\rho\mathrm{Arw}(\boldsymbol{v}_i),$$

which implies that

$$\nabla_{\boldsymbol{x}_i}\boldsymbol{u}_i = -\rho\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{v}_i).$$

By the definitions of $\boldsymbol{z}_i$ and $\boldsymbol{y}_i$, we have

$$\begin{aligned} \nabla_{\boldsymbol{x}_i}\boldsymbol{z}_i &= \frac{1}{2\rho}\Big(-\rho\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{v}_i) + \rho I\Big) \\ &= \frac{1}{2}\Big(\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(-\boldsymbol{v}_i + \boldsymbol{u}_i)\Big) \\ &= \rho\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{z}_i), \end{aligned} \tag{24}$$

$$\begin{aligned} \nabla_{\boldsymbol{x}_i}\boldsymbol{y}_i &= \frac{1}{2\rho}\Big(-\rho\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{v}_i) - \rho I\Big) \\ &= \frac{1}{2}\Big(\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(-\boldsymbol{v}_i - \boldsymbol{u}_i)\Big) \\ &= -\rho\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{y}_i). \end{aligned} \tag{25}$$

In a similar way, we can prove

$$\begin{aligned} \nabla_{\boldsymbol{s}_i}\boldsymbol{u}_i &= \mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{v}_i), \\ \frac{\partial \boldsymbol{u}_i}{\partial \mu} &= 2\rho\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\boldsymbol{e}_i. \end{aligned} \tag{26}$$

Then, we get

$$\begin{aligned} \nabla_{\boldsymbol{s}_i}\boldsymbol{z}_i &= \frac{1}{2\rho}\Big(\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{v}_i) - I\Big) \\ &= \frac{1}{2\rho}\Big(\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{v}_i - \boldsymbol{u}_i)\Big) \\ &= -\mathrm{Arw}^{-1}(\boldsymbol{u}_i)\,\mathrm{Arw}(\boldsymbol{z}_i), \end{aligned} \tag{27}$$

$$\begin{aligned}\nabla_{s_i}\boldsymbol{y}_i &= \frac{1}{2\rho}\left(\operatorname{Arw}^{-1}(\boldsymbol{u}_i)\operatorname{Arw}(\boldsymbol{v}_i)+I\right)\\ &= \frac{1}{2\rho}\left(\operatorname{Arw}^{-1}(\boldsymbol{u}_i)\operatorname{Arw}(\boldsymbol{v}_i+\boldsymbol{u}_i)\right)\\ &= \operatorname{Arw}^{-1}(\boldsymbol{u}_i)\operatorname{Arw}(\boldsymbol{y}_i),\\ \frac{\partial \boldsymbol{z}_i}{\partial\mu} &= \frac{\partial \boldsymbol{y}_i}{\partial\mu} = \operatorname{Arw}^{-1}(\boldsymbol{u}_i)\,\boldsymbol{e}_i,\end{aligned}\tag{28}$$

which completes our proof.

The following lemma gives an equivalence of the second-order cone constraints and the complementary slackness conditions of problem (1).

Lemma 4 *Given $\mu \geqslant 0$, then for any $i = 1, \cdots, n$,*

$$\boldsymbol{z}_i(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho) - \boldsymbol{x}_i = 0 \Longleftrightarrow \boldsymbol{x}_i \in K_i,\ \boldsymbol{s}_i \in K_i,\ \boldsymbol{x}_i \circ \boldsymbol{s}_i = \mu \boldsymbol{e}_i. \tag{29}$$

Proof " $\Longleftarrow$ ": Suppose $\boldsymbol{x}_i \in K_i,\ \boldsymbol{s}_i \in K_i,\ \boldsymbol{x}_i \circ \boldsymbol{s}_i = \mu\boldsymbol{e}_i$, which together with (8) gives

$$\begin{aligned}&(\boldsymbol{s}_i - \rho\boldsymbol{x}_i)^2 + 4\rho\mu\boldsymbol{e}_i\\ &\quad = \boldsymbol{s}_i^2 - 2\rho\boldsymbol{x}_i\circ\boldsymbol{s}_i + \rho^2\boldsymbol{x}_i^2 + 4\rho\mu\boldsymbol{e}_i\\ &\quad = \boldsymbol{s}_i^2 + 2\rho\boldsymbol{x}_i\circ\boldsymbol{s}_i + \rho^2\boldsymbol{x}_i^2\\ &\quad = (\boldsymbol{s}_i + \rho\boldsymbol{x}_i)^2.\end{aligned}$$

Therefore, we obtain

$$\boldsymbol{z}_i(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho) - \boldsymbol{x}_i = \frac{\left((\boldsymbol{s}_i - \rho\boldsymbol{x}_i)^2 + 4\rho\mu\boldsymbol{e}_i\right)^{\frac{1}{2}} - (\boldsymbol{s}_i + \rho\boldsymbol{x}_i)}{2\rho} = 0.$$

" $\Longrightarrow$ ": Suppose $\boldsymbol{z}_i(\boldsymbol{x}_i, \boldsymbol{s}_i; \mu, \rho) - \boldsymbol{x}_i = 0$, we have $\left((\boldsymbol{s}_i - \rho\boldsymbol{x}_i)^2 + 4\rho\mu\boldsymbol{e}_i\right)^{\frac{1}{2}} = (\boldsymbol{s}_i + \rho\boldsymbol{x}_i)$, which implies that $\boldsymbol{u}_i = \boldsymbol{s}_i + \rho\boldsymbol{x}_i$. After simplifying, we get $\boldsymbol{x}_i \circ \boldsymbol{s}_i = \mu\boldsymbol{e}_i$.

Given the spectral decomposition $\boldsymbol{v}_i = \eta_1\boldsymbol{d}_1 + \eta_2\boldsymbol{d}_2$, one can deduce

$$\begin{aligned}\boldsymbol{u}_i &= \left((\boldsymbol{s}_i - \rho\boldsymbol{x}_i)^2 + 4\rho\mu\boldsymbol{e}_i\right)^{\frac{1}{2}}\\ &= \left(\eta_1^2\boldsymbol{d}_1 + \eta_2^2\boldsymbol{d}_2 + 4\rho\mu\boldsymbol{d}_1 + 4\rho\mu\boldsymbol{d}_2\right)^{\frac{1}{2}}\\ &= \sqrt{\eta_1^2 + 4\rho\mu}\,\boldsymbol{d}_1 + \sqrt{\eta_2^2 + 4\rho\mu}\,\boldsymbol{d}_2.\end{aligned}$$

Since $\sqrt{\eta_1^2 + 4\rho\mu} \geqslant 0$ and $\sqrt{\eta_2^2 + 4\rho\mu} \geqslant 0$, we have $\boldsymbol{u_i} \in K_i$. The decomposition of $\boldsymbol{u_i}$ and $\boldsymbol{v}_i$ gives

$$\begin{aligned} \boldsymbol{s}_i &= \frac{\boldsymbol{v}_i + \boldsymbol{u}_i}{2} = \frac{\sqrt{\eta_1^2 + 4\rho\mu} + \eta_1}{2}\boldsymbol{d}_1 + \frac{\sqrt{\eta_2^2 + 4\rho\mu} + \eta_2}{2}\boldsymbol{d}_2, \\ \boldsymbol{x}_i &= \frac{\boldsymbol{u}_i - \boldsymbol{v}_i}{2\rho} = \frac{\sqrt{\eta_1^2 + 4\rho\mu} - \eta_1}{2\rho}\boldsymbol{d}_1 + \frac{\sqrt{\eta_2^2 + 4\rho\mu} - \eta_2}{2\rho}\boldsymbol{d}_2. \end{aligned} \tag{30}$$

From (30), we can see that the eigenvalues of $\boldsymbol{x}_i$ and $\boldsymbol{s}_i$ are nonnegative; thus, $\boldsymbol{x}_i,\ \boldsymbol{s}_i \in K_i$.

Given Lemma 4, the following important conclusion can be drawn.

Theorem 3 *$\boldsymbol{x}^*$ is an optimal solution of problem (1) if and only if there exist $\boldsymbol{\lambda}^* \in \mathbb{R}^m$ and $\boldsymbol{s}^* \in \mathbb{R}^k$ such that $\left(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{s}^*\right)$ is a solution of the following system:*

$$\mu = 0, \tag{31a}$$
$$A\boldsymbol{x} = \boldsymbol{b}, \tag{31b}$$
$$A^\top\boldsymbol{\lambda} + \boldsymbol{s} = \boldsymbol{c}, \tag{31c}$$
$$\boldsymbol{z}\left(\boldsymbol{x}, \boldsymbol{s}; \mu, \rho\right) - \boldsymbol{x} = \boldsymbol{0}. \tag{31d}$$

Proof The optimal condition of problem (1) is

$$A\boldsymbol{x} = \boldsymbol{b}, \tag{32a}$$
$$A^\top\boldsymbol{\lambda} + \boldsymbol{s} = \boldsymbol{c}, \tag{32b}$$
$$\boldsymbol{x}_i \circ \boldsymbol{s}_i = \boldsymbol{0},\ \boldsymbol{x}_i \in K_i,\ \boldsymbol{s}_i \in K_i,\ i = 1, \cdots, n. \tag{32c}$$

Then, the desired conclusion is directly deduced from Lemma 4.

Theorem 3 presents a novel optimal condition for the SOCP problems. The second-order cone constraints and the complementary slackness conditions can be replaced by $\boldsymbol{z}(\boldsymbol{x}, \boldsymbol{s}; \mu, \rho) - \boldsymbol{x} = \boldsymbol{0}$. These nonlinear equations are also known as the relaxed inequality constraints.

However, there is no guarantee that $\boldsymbol{u}_i \in \mathrm{int}(K_i)$, $i = 1, \cdots, n$ when $\mu = 0$, so the conclusions in Lemma 3 cannot be applied directly. As a result, the smoothing functions for (31) must be constructed. We define $\boldsymbol{\xi}_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s}) = \boldsymbol{z}(\boldsymbol{x}, \boldsymbol{s}; \mu, \rho) - \boldsymbol{x}$ and the merit function $\phi_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s})$ (ϕ in short) as follows:

$$\phi_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s}) = \frac{1}{2}\left\|\boldsymbol{\xi}_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s})\right\|^2. \tag{33}$$

Based on the newly introduced merit function ϕ, the smoothing functions of (31) can be defined as

$$\mu + \gamma\phi_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s}) = 0, \tag{34a}$$
$$A\boldsymbol{x} = \boldsymbol{b}, \tag{34b}$$
$$A^\top\boldsymbol{\lambda} + \boldsymbol{s} = \boldsymbol{c}, \tag{34c}$$
$$\boldsymbol{\xi}_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s}) = \boldsymbol{0}, \tag{34d}$$

where the parameter $\gamma > 0$ with its specific form to be provided in Algorithm 1 in the next section. It should be noted that the $\mu = 0$ in (31) has been replaced by $\mu+\gamma\phi = 0$ in (34). Since both μ and ϕ are nonnegative, $\mu + \gamma\phi = 0$ implies that $\mu = 0$ and $\phi = 0$.

4 A Primal-Dual Interior-Point Relaxation Algorithm

Based on the SBAL function, we design a primal-dual interior-point relaxation algorithm in this section. Particularly, we provide an explicit form of the Schur complement matrix in this algorithm. The explicit form contains a low-rank structure, which is utilized to make the matrix factorization more efficient and stable.

We use Newton's method to solve (34) and obtain

$$\Delta\mu = -\mu + \gamma\phi, \tag{35a}$$
$$A\Delta\boldsymbol{x} = \boldsymbol{r_p} := \boldsymbol{b} - A\boldsymbol{x}, \tag{35b}$$
$$A^\top\Delta\boldsymbol{\lambda} + \Delta\boldsymbol{s} = \boldsymbol{r_d} := \boldsymbol{c} - A^\top\boldsymbol{\lambda} - \boldsymbol{s}, \tag{35c}$$
$$\mathrm{Arw}(\rho\boldsymbol{y})\Delta\boldsymbol{x} + \mathrm{Arw}(\boldsymbol{z})\Delta\boldsymbol{s} = \boldsymbol{r_c} := \mathrm{Arw}(\boldsymbol{u})(\boldsymbol{z} - \boldsymbol{x}) + \Delta\mu\boldsymbol{e}. \tag{35d}$$

***Remark* 1** Notice that

$$\nabla\phi_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s}) = \nabla\boldsymbol{\xi}_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s})\boldsymbol{\xi}_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s})$$

and

$$\nabla\boldsymbol{\xi}_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s})^\top\boldsymbol{d} = -\boldsymbol{\xi}_{(\mu,\rho)}(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{s}),$$

where $\boldsymbol{d} = (\Delta\mu, \Delta\boldsymbol{x}, \Delta\boldsymbol{\lambda}, \Delta\boldsymbol{s})$. Thus, it can be deduced that $\nabla\phi^\top\boldsymbol{d} = -2\phi$. By combining with Newton's equation $\nabla\mu + \gamma\nabla\phi^\top\boldsymbol{d} = -\mu - \gamma\phi$, Eq. (35a) can be derived. Equation (35d) is obtained from Lemma 3.

Lemma 2 shows that $\mathrm{Arw}(\rho\boldsymbol{y})$ and $\mathrm{Arw}(\boldsymbol{z})$ are commutative in our algorithm. The matrices corresponding to the HKM direction and NT direction in the interior-point methods are also commutative. The details can be found in [1].

Lemma 5 *If $\mu > 0$, then $Arw^{-1}(\rho \boldsymbol{y})Arw(\boldsymbol{z})$ and $Arw(\boldsymbol{z})Arw^{-1}(\rho \boldsymbol{y})$ are symmetric positive definite.*

Proof By the first conclusion of Lemma 2, we have

$$\mathrm{Arw}(\boldsymbol{z})\mathrm{Arw}(\rho \boldsymbol{y}) = \mathrm{Arw}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z}),$$

which implies that

$$\mathrm{Arw}^{-1}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z}) = \mathrm{Arw}(\boldsymbol{z})\mathrm{Arw}^{-1}(\rho \boldsymbol{y}). \tag{36}$$

Notice that

$$\begin{aligned}(\mathrm{Arw}^{-1}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z}))^\top &= \mathrm{Arw}(\boldsymbol{z})^\top \mathrm{Arw}^{-\top}(\rho \boldsymbol{y}) = \mathrm{Arw}(\boldsymbol{z})\mathrm{Arw}^{-1}(\rho \boldsymbol{y})\\ &= \mathrm{Arw}^{-1}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z}).\end{aligned}$$

Hence, $\mathrm{Arw}^{-1}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})$ is symmetric. Similarly, we can prove $\mathrm{Arw}(\boldsymbol{z})\mathrm{Arw}^{-1}(\rho \boldsymbol{y})$ is symmetric.

If $\mu > 0$, then the spectral values of both $\boldsymbol{z}$ and $\boldsymbol{y}$ are greater than zero. Thus, we have $\boldsymbol{z} \in \mathrm{int}(K)$, $\rho \boldsymbol{y} \in \mathrm{int}(K)$. Observe that any vector $\boldsymbol{x} \in K$ ($\boldsymbol{x} \in \mathrm{int}(K)$) if and only if $\mathrm{Arw}(\boldsymbol{x})$ is positive semidefinite (positive definite). This is because $\mathrm{Arw}(\boldsymbol{x})$ is positive semidefinite if and only if either $\boldsymbol{x} = \boldsymbol{0}$, or $x_0 > 0$ and the Schur complement $x_0 - \bar{\boldsymbol{x}}^\top (x_0 I)^{-1}\bar{\boldsymbol{x}} \geqslant 0$. So $\mathrm{Arw}(\boldsymbol{z})$ and $\mathrm{Arw}(\rho \boldsymbol{y})$ are positive definite, which also shows that $\mathrm{Arw}^{-1}(\boldsymbol{z})$ and $\mathrm{Arw}^{-1}(\rho \boldsymbol{y})$ are positive definite. Equation (36) implies that $\mathrm{Arw}(\boldsymbol{z})$ and $\mathrm{Arw}^{-1}(\rho \boldsymbol{y})$ are commutative, therefore, $\mathrm{Arw}^{-1}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})$ and $\mathrm{Arw}(\boldsymbol{z})\mathrm{Arw}^{-1}(\rho \boldsymbol{y})$ are positive definite.s

Theorem 4 *If $\mu > 0$ and A has full row rank, then (35) has one unique solution $(\Delta \boldsymbol{x}; \Delta \boldsymbol{\lambda}; \Delta \boldsymbol{s}; \Delta \mu) \in \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R}$.*

Proof It is worth noting that $\Delta\mu$ already has an explicit solution. All that remains is to demonstrate that the following system

$$A\Delta \boldsymbol{x} = \boldsymbol{0}, \tag{37a}$$

$$A^\top \Delta \boldsymbol{\lambda} + \Delta \boldsymbol{s} = \boldsymbol{0}, \tag{37b}$$

$$\mathrm{Arw}(\rho \boldsymbol{y})\Delta \boldsymbol{x} + \mathrm{Arw}(\boldsymbol{z})\Delta \boldsymbol{s} = \boldsymbol{0}, \tag{37c}$$

only has a zero solution, i.e., $(\Delta \boldsymbol{x}; \Delta \boldsymbol{\lambda}; \Delta \boldsymbol{s}) = (\boldsymbol{0}; \boldsymbol{0}; \boldsymbol{0})$.

Equation (37c) implies that $\Delta \boldsymbol{x} = -\mathrm{Arw}^{-1}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})\Delta \boldsymbol{s}$, which together with (37b) gives

$$0 = \Delta \boldsymbol{x}^\top A^\top \Delta \boldsymbol{\lambda} = -\Delta \boldsymbol{x}^\top \Delta \boldsymbol{s} = \Delta \boldsymbol{s}^\top \mathrm{Arw}(\boldsymbol{z})\mathrm{Arw}^{-1}(\rho \boldsymbol{y})\Delta \boldsymbol{s}. \tag{38}$$

According to Lemma 5, $\mathrm{Arw}(\boldsymbol{z})\mathrm{Arw}^{-1}(\rho \boldsymbol{y})$ is positive definite, thus implying $\Delta \boldsymbol{s} = \boldsymbol{0}$. Equations (37b) and (37c) yield $\Delta \boldsymbol{x} = -\mathrm{Arw}^{-1}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})\Delta \boldsymbol{s} = \boldsymbol{0}$ and $A^\top \Delta \boldsymbol{\lambda} = \boldsymbol{0}$. Since A has full row rank, we obtain $\Delta \boldsymbol{\lambda} = \boldsymbol{0}$.

By substituting $\Delta \boldsymbol{s} = \mathrm{Arw}^{-1}(\boldsymbol{z})\,(\boldsymbol{r_c} - \mathrm{Arw}(\rho\boldsymbol{y})\Delta\boldsymbol{x})$ into (35c), the augmented system can be obtained:

$$\Delta\mu = -\mu + \gamma\phi, \tag{39a}$$
$$\begin{pmatrix} -\mathrm{Arw}^{-1}(\boldsymbol{z})\mathrm{Arw}(\rho\boldsymbol{y}) & A^\top \\ A & \boldsymbol{0} \end{pmatrix}\begin{pmatrix} \Delta\boldsymbol{x} \\ \Delta\boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} \boldsymbol{r_d} - \mathrm{Arw}^{-1}(\boldsymbol{z})\boldsymbol{r_c} \\ \boldsymbol{r_p} \end{pmatrix}, \tag{39b}$$
$$\Delta\boldsymbol{s} = \boldsymbol{r_d} - A^\top\Delta\boldsymbol{\lambda}. \tag{39c}$$

After further simplification, (35) can be transformed into the following Schur complement system:

$$\Delta\mu = -\mu + \gamma\phi, \tag{40a}$$
$$A\mathrm{Arw}^{-1}(\rho\boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})A^\top\Delta\boldsymbol{\lambda} = \boldsymbol{r_p} + A\mathrm{Arw}^{-1}(\rho\boldsymbol{y})\left(\mathrm{Arw}(\boldsymbol{z})\boldsymbol{r_d} - \boldsymbol{r_c}\right), \tag{40b}$$
$$\Delta\boldsymbol{s} = \boldsymbol{r_d} - A^\top\Delta\boldsymbol{\lambda}, \tag{40c}$$
$$\Delta\boldsymbol{x} = \mathrm{Arw}^{-1}(\rho\boldsymbol{y})\left(\boldsymbol{r_c} - \mathrm{Arw}(\boldsymbol{z})\Delta\boldsymbol{s}\right). \tag{40d}$$

The most time-consuming processes of solving (40) are the formation and factorization of the Schur complement matrix, $A\mathrm{Arw}^{-1}(\rho\boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})A^\top$. Even if A is sparse, the Schur complement matrix could be dense, making it difficult to solve. To improve the efficiency of calculations, we analyze the form of $G := \mathrm{Arw}^{-1}(\rho\boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})$. [17] gives some computational details of $AGA^\top$, but G in [17] is calculated directly. In the following, we will give the explicit form of G. To the best of our knowledge, there are no such results for non-interior-point methods. Firstly, we consider the case of a single cone.

Lemma 6 *Suppose that $\boldsymbol{x},\ \boldsymbol{s} \in \mathbb{R}^k$. Let $\boldsymbol{z}$ and $\boldsymbol{y}$ be defined by (16) and (17), respectively. Then,*

$$G = Arw^{-1}(\rho\boldsymbol{y})Arw(\boldsymbol{z}) = \frac{1}{\mu}\left(\det(\boldsymbol{z})I + 2\boldsymbol{z}\boldsymbol{z}^\top - 2\det(\boldsymbol{z})\boldsymbol{e}\boldsymbol{e}^\top\right).$$

Proof Notice that $\rho\boldsymbol{y}\circ\boldsymbol{z} = \mu\boldsymbol{e}$ is equivalent to

$$z_0y_0 + \bar{\boldsymbol{z}}^\top\bar{\boldsymbol{y}} = \frac{\mu}{\rho},$$
$$z_0\bar{\boldsymbol{y}} + y_0\bar{\boldsymbol{z}} = 0,$$

which leads to $\det(\boldsymbol{z}) = \mu\dfrac{z_0}{\rho y_0}$.

Based on Lemma 33 in [1] and the fact that $\rho \boldsymbol{y} \circ \boldsymbol{z} = \mu \boldsymbol{e}$, the following equation can be obtained:

$$\begin{aligned} G &= \mathrm{Arw}^{-1}(\rho \boldsymbol{y})\mathrm{Arw}(\boldsymbol{z}) \\ &= \beta I - \gamma R\boldsymbol{z}\boldsymbol{e}^\top + \gamma R\rho \boldsymbol{y}\left(\boldsymbol{z} \circ (\rho \boldsymbol{y})^{-1}\right)^\top \\ &= \beta I - \gamma R\boldsymbol{z}\boldsymbol{e}^\top + \gamma R\boldsymbol{z}^{-1}\left(\boldsymbol{z}^2\right)^\top, \end{aligned}$$

where $\gamma = \dfrac{1}{\rho y_0}$, $\beta = \dfrac{\det(\boldsymbol{z})}{\mu}$. Note that

$$\begin{aligned} \boldsymbol{z}^{-1}\left(\boldsymbol{z}^2\right)^\top &= \frac{1}{\det(\boldsymbol{z})} R \begin{pmatrix} z_0 \\ \bar{\boldsymbol{z}} \end{pmatrix} \left(\|\boldsymbol{z}\|^2 \;\; 2z_0 \bar{\boldsymbol{z}}^\top \right) \\ &= \frac{1}{\det(\boldsymbol{z})} R \begin{pmatrix} z_0\|\boldsymbol{z}\|^2 & 2z_0^2\bar{\boldsymbol{z}}^\top \\ \|\boldsymbol{z}\|^2\bar{\boldsymbol{z}} & 2z_0\bar{\boldsymbol{z}}\bar{\boldsymbol{z}}^\top \end{pmatrix}. \end{aligned} \tag{41}$$

Therefore,

$$\begin{aligned} &-\gamma R\boldsymbol{z}\boldsymbol{e}^\top + \gamma R\boldsymbol{z}^{-1}\left(\boldsymbol{z}^2\right)^\top \\ &= -\gamma R\boldsymbol{z}\boldsymbol{e}^\top + \frac{\gamma}{\det(\boldsymbol{z})} \begin{pmatrix} z_0\|\boldsymbol{z}\|^2 & 2z_0^2\bar{\boldsymbol{z}}^\top \\ \|\boldsymbol{z}\|^2\bar{\boldsymbol{z}} & 2z_0\bar{\boldsymbol{z}}\bar{\boldsymbol{z}}^\top \end{pmatrix} \\ &= \frac{\gamma}{\det(\boldsymbol{z})} \begin{pmatrix} z_0\|\boldsymbol{z}\|^2 - z_0\det(\boldsymbol{z}) & 2z_0\bar{\boldsymbol{z}}^\top \\ \|\boldsymbol{z}\|^2\bar{\boldsymbol{z}} + \det(\boldsymbol{z})\bar{\boldsymbol{z}} & 2z_0\bar{\boldsymbol{z}}\bar{\boldsymbol{z}}^\top \end{pmatrix} \\ &= \frac{2z_0\gamma}{\det(\boldsymbol{z})} \begin{pmatrix} \|\bar{\boldsymbol{z}}\|^2 & z_0\bar{\boldsymbol{z}}^\top \\ z_0\bar{\boldsymbol{z}} & \bar{\boldsymbol{z}}\bar{\boldsymbol{z}}^\top \end{pmatrix} \\ &= \frac{2}{\mu}\left(\boldsymbol{z}\boldsymbol{z}^\top - \begin{pmatrix} \det(\boldsymbol{z}) & 0 \\ 0 & 0 \end{pmatrix}\right) \\ &= \frac{2}{\mu}\boldsymbol{z}\boldsymbol{z}^\top - \frac{2\det(\boldsymbol{z})}{\mu}\boldsymbol{e}\boldsymbol{e}^\top. \end{aligned} \tag{42}$$

So we get $G = \dfrac{\det(\boldsymbol{z})}{\mu} I + \dfrac{2}{\mu}\boldsymbol{z}\boldsymbol{z}^\top - \dfrac{2\det(\boldsymbol{z})}{\mu}\boldsymbol{e}\boldsymbol{e}^\top$, which completes the proof.

For the case of multiple cones, by combining with Lemma 6, we have

$$
\begin{aligned}
\mathrm{AGA}^\top &= \sum_{i=1}^{n} A_i \mathrm{Arw}^{-1}(\rho \boldsymbol{y}_i)\, \mathrm{Arw}(\boldsymbol{z}_i)\, A_i^\top \\
&= \sum_{i=1}^{n} A_i \left(\frac{\det(\boldsymbol{z}_i)}{\mu} I + \frac{2}{\mu} \boldsymbol{z}_i \boldsymbol{z}_i^\top - \frac{2\det(\boldsymbol{z}_i)}{\mu} \boldsymbol{e_i} \boldsymbol{e_i}^\top \right) A_i^\top \\
&= \frac{1}{\mu} \sum_{i=1}^{n} \left(\det(\boldsymbol{z}_i) A_i A_i^\top + 2 A_i \boldsymbol{z}_i \,(A_i \boldsymbol{z}_i)^\top - 2\det(\boldsymbol{z}_i) A_i \boldsymbol{e_i}\, (A_i \boldsymbol{e_i})^\top \right) \\
&= \frac{1}{\mu} \left(M + \sum_{i=1}^{n} 2 \boldsymbol{h}_i \boldsymbol{h}_i^\top - \sum_{i=1}^{n} 2\det(\boldsymbol{z}_i) \boldsymbol{a}_i \boldsymbol{a}_i^\top \right),
\end{aligned}
\tag{43}
$$

where $M = \sum_{i=1}^{n} \det(\boldsymbol{z}_i) A_i A_i^\top$, $\boldsymbol{h}_i = A_i \boldsymbol{z}_i$, $\boldsymbol{a}_i = A_i \boldsymbol{e_i}$, $i = 1, \cdots, n$.

Even though the matrix A is sparse, the vectors $\boldsymbol{h}_i$, $i = 1, \cdots, n$ are usually dense, causing the matrix $\mathrm{AGA}^\top$ to be dense. However, we can use the sparse Cholesky factorization by defining the expanded sparse representation of $\mathrm{AGA}^\top$ shown below.

Definition 1 Define $H = (\boldsymbol{h}_1, \boldsymbol{h}_2, \cdots, \boldsymbol{h}_n)$, $\tilde{A} = (\boldsymbol{a}_1, \boldsymbol{a}_2, \cdots, \boldsymbol{a}_n)$ and two diagonal matrices $\tilde{D} = \mathrm{diag}\left(\frac{1}{2}, \frac{1}{2}, \ldots, \frac{1}{2}\right)$, $\hat{D} = \mathrm{diag}\left(\frac{1}{2\det(\boldsymbol{z}_1)}, \frac{1}{2\det(\boldsymbol{z}_2)}, \ldots, \frac{1}{2\det(\boldsymbol{z}_n)}\right)$. Then, the matrix

$$
E := \begin{pmatrix} M & \tilde{A} & H \\ \tilde{A}^\top & \hat{D} & 0 \\ H^\top & 0 & -\tilde{D} \end{pmatrix}
$$

is called the expanded sparse representation of $\mathrm{AGA}^\top$.

Instead of solving (40b) directly, we solve the following equation using the expanded sparse representation of $\mathrm{AGA}^\top$:

$$
\begin{pmatrix} M & \tilde{A} & H \\ \tilde{A}^\top & \hat{D} & 0 \\ H^\top & 0 & -\tilde{D} \end{pmatrix}
\begin{pmatrix} \Delta \boldsymbol{x} \\ \boldsymbol{q} \\ \boldsymbol{t} \end{pmatrix}
= \begin{pmatrix} \mu \left(\boldsymbol{r_p} + A \mathrm{Arw}^{-1}(\rho \boldsymbol{y}) \left(\mathrm{Arw}(\boldsymbol{z}) \boldsymbol{r_d} - \boldsymbol{r_c} \right) \right) \\ \boldsymbol{0} \\ \boldsymbol{0} \end{pmatrix},
\tag{44}
$$

where $\boldsymbol{q} \in \mathbb{R}^n$ and $\boldsymbol{t} \in \mathbb{R}^n$ are newly added variables.

***Remark* 2** One advantage of (44) over (40b) is that it prevents the construction of dense matrices, enhancing the effectiveness of the algorithm. Furthermore, according to Lemma 6, the implicit barrier parameter μ in $\mathrm{AGA}^\top$ can be extracted, which improves the stability of sparse Cholesky factorization. We consider the case when the optimal solution (x^*, s^*) satisfies $\det(x_i^*) > 0$, $\det(s_i^*) = 0$ for some $i \in \{1, 2, \cdots, n\}$. In this case, we have $\det(z_i^*) = \Theta(1)$, implying the eigenvalue of $\det(z_i^*)A_iA_i^\top$ is $\Theta(1)$. But in the NT direction, the eigenvalues of the corresponding matrix are $\Theta(\mu^{-1})$ (see [19]). Therefore, compared with interior-point methods, IPRSOCP is capable of escaping from the ill conditioning trap when $\mu \to 0$.

After obtaining the search direction via Newton's method, we need to calculate the step length. For simplicity, denote $\boldsymbol{w}^{(k)} = \left(\boldsymbol{x}^{(k)}; \boldsymbol{\lambda}^{(k)}; \boldsymbol{s}^{(k)}\right)$, $\Delta\boldsymbol{w}^{(k)} = \left(\Delta\boldsymbol{x}^{(k)}; \Delta\boldsymbol{\lambda}^{(k)}; \Delta\boldsymbol{s}^{(k)}\right)$. The Armijo line search procedure is applied to calculating step length. At the k-th iteration, we pick $\alpha \in (0, 1]$ to be the largest value in the set $\left\{1, \delta, \delta^2, \cdots\right\}$ such that the following inequality holds:

$$\phi_{(\mu^{(k)}+\alpha\Delta\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)} + \alpha\Delta\boldsymbol{w}^{(k)}\right) \leqslant (1-2\tau\alpha)\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^k), \tag{45}$$

where $0 < \delta < 1$ and $0 < \tau < 0.5$.

The merit function ϕ only considers the residuals of the relaxed inequality constraints while not taking into account the primal and dual infeasibilities. In order to detect whether the problem is convergent, we introduce the following function:

$$\psi_{(\mu,\rho)}(\boldsymbol{w}) = \frac{1}{2}\left(\left\|\boldsymbol{c} - A^\top\boldsymbol{\lambda} - \boldsymbol{s}\right\|^2 + \|A\boldsymbol{x} - \boldsymbol{b}\|^2\right) + \phi_{(\mu,\rho)}(\boldsymbol{w}). \tag{46}$$

It is worth noting that

$$\begin{aligned}\left\|A\boldsymbol{x}^{(k+1)} - \boldsymbol{b}\right\|^2 &= \left\|A\boldsymbol{x}^{(k)} - \boldsymbol{b} + \alpha^{(k)}A\Delta\boldsymbol{x}\right\|^2 = \left(1-\alpha^{(k)}\right)^2\left\|A\boldsymbol{x}^{(k)} - \boldsymbol{b}\right\|^2 \\ &\leqslant \left(1 - 2\tau\alpha^{(k)}\right)\left\|A\boldsymbol{x}^{(k)} - \boldsymbol{b}\right\|^2.\end{aligned} \tag{47}$$

By the same way, we can get

$$\left\|\boldsymbol{c} - A^\top\boldsymbol{\lambda}^{(k+1)} - \boldsymbol{s}^{(k+1)}\right\|^2 \leqslant (1-2\tau\alpha^{(k)})\left\|\boldsymbol{c} - A^\top\boldsymbol{\lambda}^{(k)} - \boldsymbol{s}^{(k)}\right\|^2. \tag{48}$$

Thus, if we choose the step length $\alpha^{(k)}$ to satisfy (45), the following inequality also holds:

$$\psi_{(\mu^{(k)}+\alpha^{(k)}\Delta\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)} + \alpha^{(k)}\Delta\boldsymbol{w}^{(k)}\right) \leqslant (1-2\tau\alpha^{(k)})\psi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^k). \tag{49}$$

With the above preparation, the primal-dual interior-point relaxation algorithm (IPRSOCP) for solving second-order cone programming problems can be presented as follows:

***Remark* 3** It is important to note that any point can be used as a starting point in Algorithm 1. The reason is that according to Theorem 3, the iterative points of IPRSOCP do not have to be inside the second-order cones.

Algorithm 1 The IPRSOCP algorithm

Given $\left(\boldsymbol{x}^{(0)}; \boldsymbol{\lambda}^{(0)}; \boldsymbol{s}^{(0)}\right) \in \mathbb{R}^k \times \mathbb{R}^m \times \mathbb{R}^k$, $\mu^{(0)} > 0$, $\rho^{(0)} > 0$, $\eta > 1$, $\gamma_0, \delta, \sigma \in (0, 1)$, $\tau \in (0, 0.5)$, $\epsilon \in \left(0, \mu^{(0)}\right)$. Calculate $\boldsymbol{z}^{(0)}$, $\boldsymbol{y}^{(0)}$ and $\phi_{\left(\mu^{(0)}, \rho^{(0)}\right)}\left(\boldsymbol{w}^{(0)}\right)$. Set $k := 0$.
Set $\gamma = \min\left\{\gamma_0,\ \mu^{(0)}/\phi_{\left(\mu^{(0)}, \rho^{(0)}\right)}\left(\boldsymbol{w}^{(0)}\right)\right\}$.
while $\mu^{(k)} \geqslant \epsilon$ or $\psi_{\left(\mu^{(k)}, \rho^{(k)}\right)}\left(\boldsymbol{w}^{(k)}\right) \geqslant \epsilon$ **do**
  Solve (40) to get the search direction $\Delta\boldsymbol{w}^{(k)}$.
  Find step length $\alpha^{(k)} \in (0, 1]$, namely search in the set $\left\{1, \delta, \delta^2, \ldots\right\}$, finding the largest value satisfying inequality (45).
  Update $\mu^{(k+1)} = \mu^{(k)} + \alpha^{(k)}\Delta\mu^{(k)}$, $\boldsymbol{w}^{(k+1)} = \boldsymbol{w}^{(k)} + \alpha^{(k)}\Delta\boldsymbol{w}^{(k)}$.
  Update $\rho^{(k+1)} = \max\left\{\rho^{(k)},\ \sigma\dfrac{\left\|s^{(k+1)}\right\|}{\max\left(1, \left\|x^{(k+1)}\right\|\right)}\right\}$.
  Calculate $\boldsymbol{z}^{(k+1)}$, $\boldsymbol{y}^{(k+1)}$ and $\phi_{\left(\mu^{(k+1)}, \rho^{(k+1)}\right)}\left(\boldsymbol{w}^{(k+1)}\right)$.
  Set $\gamma = \min\left\{\gamma_0,\ \mu^{(k+1)}/\phi_{\left(\mu^{(k+1)}, \rho^{(k+1)}\right)}\left(\boldsymbol{w}^{(k+1)}\right)\right\}$.
  Set $k := k + 1$
end while

Theorem 5 ensures that if the merit function ϕ is not zero, we can always find an appropriate step length $\alpha^{(k)}$ to continue the algorithm. Thus, Algorithm 1 is well defined.

Theorem 5 *There always exists a step length* $\alpha^{(k)} \in (0, 1]$ *satisfying the inequality* (45).

Proof The proof can be found in Appendix A.

The iterative point $\left(\mu^{(k)}, \boldsymbol{w}^{(k)}\right)$ falls into the following set when $\gamma = \gamma_0$:

$$N(\gamma_0) := \left\{\left(\mu^{(k)}; \boldsymbol{w}^{(k)}\right) : \gamma_0\phi_{\left(\mu^{(k)}, \rho^{(k)}\right)}\left(\boldsymbol{w}^{(k)}\right) \leqslant \mu^{(k)}\right\}. \tag{50}$$

$N(\gamma_0)$ defines a neighborhood of the central path, and a similar definition for linear programming can be found in [15, 48]. Figure 1 illustrates the difference between interior-point methods and IPRSOCP on the central path.

The iterative points for interior-point path-following methods are shown in the left image of Fig. 1. Because IPRSOCP allows iterative points to be outside the second-order cone, the design of central path is more flexible. On the one hand, compared

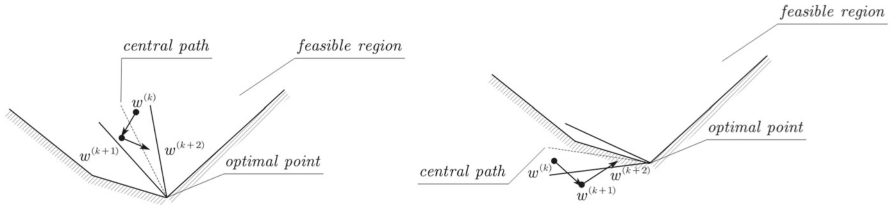

Fig. 1 Iterative points of interior-point methods and IPRSOCP.

with interior-point methods, the neighborhood of the central path in IPRSOCP can be outside of the feasible region. On the other hand, compared with the methods mentioned in [15, 48] the iterative points are allowed to be outside the neighborhood $N(\gamma_0)$. When $\left(\mu^{(k)}; \boldsymbol{w}^{(k)}\right) \notin N(\gamma_0)$, it means that the barrier parameter $\mu^{(k)}$ decreases too fast. At this point, $\mu^{(k)}$ will not be updated in accordance with the definition of γ, guaranteeing the numerical stability of the algorithm.

5 Global Convergence

In this section, the global convergence of IPRSOCP is established. The convergent results are based on the following two assumptions:

Assumption 1 The coefficient matrix A is of full row rank.

Assumption 2 The iterative sequence $\{\boldsymbol{x}^{(k)}\}$ is bounded.

Theorem 5 guarantees that the merit function ϕ decreases once the step length is selected. The following lemma ensures that when the penalty parameter ρ is adjusted, the value of ϕ decreases as well.

Lemma 7 *Given $\mu > 0$ and $\rho > 0$, then $\boldsymbol{z}_i$, $i = 1, \cdots, n$ are differentiable with respect to the penalty parameter ρ, and*

$$\frac{\partial \boldsymbol{z}_i}{\partial \rho} = Arw^{-1}(\boldsymbol{u}_i) Arw(\boldsymbol{z}_i)(\boldsymbol{x}_i - \boldsymbol{z}_i).$$

Thus,

$$\frac{\partial \|\boldsymbol{z}_i - \boldsymbol{x}_i\|^2}{\partial \rho} = -2(\boldsymbol{z}_i - \boldsymbol{x}_i)^\top Arw^{-1}(\boldsymbol{u}_i) Arw(\boldsymbol{z}_i)(\boldsymbol{z}_i - \boldsymbol{x}_i).$$

Proof Taking the derivative of $\boldsymbol{y}_i \circ \boldsymbol{z}_i = \frac{\mu}{\rho}\boldsymbol{e}_i$ and $\boldsymbol{z}_i - \boldsymbol{y}_i = \boldsymbol{x}_i - \frac{\boldsymbol{s}_i}{\rho}$ with respect to the penalty parameter ρ, we obtain

$$\begin{aligned} \boldsymbol{y}_i \circ \frac{\partial \boldsymbol{z}_i}{\partial \rho} + \frac{\partial \boldsymbol{y}_i}{\partial \rho} \circ \boldsymbol{z}_i &= -\frac{\mu}{\rho^2}\boldsymbol{e}_i = -\frac{\boldsymbol{y}_i \circ \boldsymbol{z}_i}{\rho}, \\ \frac{\partial \boldsymbol{z}_i}{\partial \rho} - \frac{\partial \boldsymbol{y}_i}{\partial \rho} &= \frac{\boldsymbol{s}_i}{\rho^2} = \frac{\boldsymbol{x}_i - (\boldsymbol{z}_i - \boldsymbol{y}_i)}{\rho}. \end{aligned}$$

After a simple calculation, we can get $\frac{\partial \boldsymbol{z}_i}{\partial \rho} = \mathrm{Arw}^{-1}(\boldsymbol{u}_i)\mathrm{Arw}(\boldsymbol{z}_i)(\boldsymbol{x}_i - \boldsymbol{z}_i)$ and

$$\frac{\partial \|\boldsymbol{z}_i - \boldsymbol{x}_i\|^2}{\partial \rho} = 2(\boldsymbol{z}_i - \boldsymbol{x}_i)^\top \mathrm{Arw}^{-1}(\boldsymbol{u}_i)\mathrm{Arw}(\boldsymbol{z}_i)(\boldsymbol{x}_i - \boldsymbol{z}_i),$$

which completes the proof.

The monotonicity of the merit function ϕ can be obtained by Lemma 7 as follows:

Lemma 8 *Denote*

$$\hat{z}^{(k+1)} = z\left(\boldsymbol{x}^{(k+1)}, \boldsymbol{s}^{(k+1)}; \mu^{(k+1)}, \rho^{(k)}\right),$$
$$\hat{\boldsymbol{u}}^{(k+1)} = \boldsymbol{u}\left(\boldsymbol{x}^{(k+1)}, \boldsymbol{s}^{(k+1)}; \mu^{(k+1)}, \rho^{(k)}\right).$$

Assuming $\|\hat{z}^{(k+1)} - \boldsymbol{x}^{(k+1)}\| \neq 0$ *and* $\phi_{(\mu^{(k+1)},\rho^{(k)})}(\boldsymbol{w}^{(k+1)}) \leqslant \phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})$*, if* $\rho^{(k+1)} \geqslant \rho^{(k)} > 0$*, then*

$$\phi_{(\mu^{(k+1)},\rho^{(k+1)})}(\boldsymbol{w}^{(k+1)}) \leqslant \phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}).$$

Proof According to Lemma 7, we have

$$\begin{aligned}
\left.\frac{\partial \phi_{(\mu^{(k+1)},\rho)}(\boldsymbol{w}^{(k+1)})}{\partial \rho}\right|_{\rho=\rho^{(k)}} &= \frac{1}{2}\sum_{i=1}^{n} \left.\frac{\partial \|z_i - \boldsymbol{x}_i\|^2}{\partial \rho}\right|_{\rho=\rho^{(k)}} \\
&= -\sum_{i=1}^{n} (\hat{z}_i - \boldsymbol{x}_i)^\top \mathrm{Arw}^{-1}(\hat{\boldsymbol{u}}_i)\mathrm{Arw}(\hat{z}_i)(\hat{z}_i - \boldsymbol{x}_i) < 0. \qquad (51)
\end{aligned}$$

The inequality (51) states that $\phi_{(\mu^{(k+1)},\rho)}(\boldsymbol{w}^{(k+1)})$ is monotonically decreasing with respect to the parameter ρ, which proves the conclusion.

Since $\{\rho^{(k)}\}$ is monotonically non-decreasing, we can conclude that the value of ϕ is monotonically non-increasing in IPRSOCP according to Lemma 8.

Therefore, $\{\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})\}$ is bounded, and $\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}) \leqslant \phi_{(\mu^{(0)},\rho^{(0)})}(\boldsymbol{w}^{(0)})$ for all $k \geqslant 0$. By the definition of γ, we have

$$\begin{aligned}
\mu^{(k+1)} &= \mu^{(k)} + \alpha^{(k)}\Delta\mu^{(k)} \\
&\leqslant \mu^{(k)} + \alpha^{(k)}(-\mu^{(k)} + \frac{\mu^{(k)}}{\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})}\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})) \\
&= \mu^{(k)},
\end{aligned}$$

which leads to the fact that $\{\mu^{(k)}\}$ is also monotonically non-increasing.

Corollary 1 *The function* $\psi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})$ *is monotonically non-increasing, that is, for* $k = 0, 1, 2, \cdots$,

$$\psi_{(\mu^{(k+1)},\rho^{(k+1)})}(\boldsymbol{w}^{(k+1)}) \leqslant \psi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}).$$

Proof This conclusion can be derived directly from inequalities (47), (48) and Lemma 8.

Theorem 6 *Under Assumption 2, $\{\boldsymbol{z}^{(k)}\}$ and $\{\boldsymbol{y}^{(k)}\}$ are bounded, and there exists a scalar $\hat{\tau} > 0$ such that the following inequalities hold for $i = 1, \cdots, n$:*

$$\|\boldsymbol{y}_i^{(k)}\| \geqslant \frac{\hat{\tau}\mu^{(k)}}{\rho^{(k)}}, \ \|\boldsymbol{z}_i^{(k)}\| \geqslant \frac{\hat{\tau}\mu^{(k)}}{\rho^{(k)}}.$$

Proof The proof can be found in Appendix A.

***Remark* 4** In terms of computation, we present the expanded sparse representation of $\mathrm{AGA}^\top$. However, we continue to analyze using the original Schur complement matrix. Suppose that the lower bound of μ is strictly greater than zero and that $\{\rho^{(k)}\}$ is bounded.

Combined with Theorem 1 and Theorem 6, we know that the eigenvalues of $\mathrm{Arw}^{-1}(\rho\boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})$ are all positive and bounded, and the minimum eigenvalue of $\mathrm{Arw}^{-1}(\rho\boldsymbol{y})\mathrm{Arw}(\boldsymbol{z})$ is strictly greater than zero. Under Assumption 1, the matrix $AGA^\top$ is non-singular.

Theorem 7 *Under Assumptions 1 and 2, if $\{\rho^{(k)}\}$ is upper bounded, then $\{\boldsymbol{s}^{(k)}\}$ and $\{\boldsymbol{\lambda}^{(k)}\}$ are also bounded.*

Proof If $\{\rho^{(k)}\}$ has an upper bound, then $\{\boldsymbol{s}^{(k)}\}$ is bounded, since

$$\|\boldsymbol{s}^{(k)}\| \leqslant \rho^{(k)} \max\{\|\boldsymbol{x}^{(k)}\|, 1\}/\sigma$$

for all $k = 1, 2, \cdots$. According to the inequality

$$\frac{1}{2}\|\boldsymbol{c} - A^\top\boldsymbol{\lambda}^{(k)} - \boldsymbol{s}^{(k)}\|^2 \leqslant \psi_{(\mu^{(0)},\rho^{(0)})}(\boldsymbol{w}^{(0)})$$

and the fact that A is of full row rank, it can be inferred that $\{\boldsymbol{\lambda}^{(k)}\}$ is bounded, which completes the proof.

Theorem 8 *Under Assumptions 1 and 2, if $\{\rho^{(k)}\}$ is upper bounded, then for $\{\mu^{(k)}\}$ and $\{\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})\}$ generated by Algorithm 1, we have*

$$\lim_{k\to\infty} \phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}) = 0, \ \lim_{k\to\infty} \mu^{(k)} = 0.$$

Proof Note that $\{\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})\}$ is a monotonically non-increasing sequence. Thus, by the boundedness of $\{\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})\}$, there exists a scalar $\phi^* \geqslant 0$ such that

$$\lim_{k\to\infty} \phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}) = \phi^*, \ \lim_{k\to\infty} \alpha^{(k)}\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}) = 0.$$

If $\phi^* > 0$, then we have $\lim_{k\to\infty} \alpha^{(k)} = 0$ and $\liminf_{k\to\infty} \mu^{(k)} > 0$ since $\mu^{(k)}$ remains unchanged, providing $\mu^{(k)} \leqslant \gamma_0\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})$. Hence, $\boldsymbol{z}^{(k)}$ and $\boldsymbol{y}^{(k)}$ are bounded away from zero by Lemma 6. Due to Remark 4 and the boundedness of the merit function, $\{\Delta \boldsymbol{w}^{(k)}\}$ is bounded. In this case, $\alpha^{(k)}$ is bounded away from zero since

$$\begin{aligned}
&\phi_{(\mu^{(k)}+\alpha\Delta\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}+\alpha\Delta\boldsymbol{w}^{(k)}) - (1-2\tau\alpha)\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})\\
&= -2(1-\tau)\alpha\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}) + o(\alpha)\\
&\leqslant -2(1-\tau)\phi^*\alpha + o(\alpha),
\end{aligned}$$

which suggests that there exists $\alpha^* \in (0, 1)$ such that (45) holds for all $\alpha \in (0, \alpha^*]$. It is contrary to $\lim_{k\to\infty} \alpha^{(k)} = 0$. This contradiction shows $\phi^* = 0$.

If there exists a scalar K such that $\mu^{(k)} > \gamma_0\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})$ for all $k > K$, then $\gamma = \gamma_0$. Suppose that $\lim_{k\to\infty} \mu^{(k)} = \mu^* > 0$. As discussed above, we can find $\alpha^* \in (0, 1)$ such that (45) holds for all $\alpha \in (0, \alpha^*]$. Therefore, we can pick $\alpha^{(k)} > \delta\alpha^*$ to satisfy the following inequality:

$$\begin{aligned}
\mu^{(k+1)} &= (1-\alpha^{(k)})\mu^{(k)} + \alpha^{(k)}\gamma_0\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})\\
&\leqslant (1-\delta\alpha^*)\mu^{(k)} + \alpha^{(k)}\gamma_0\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}).
\end{aligned}$$

By taking the limits on both sides, we can obtain

$$\mu^* \leqslant (1-\delta\alpha^*)\mu^*,$$

which implies that $\mu^* = 0$. Otherwise, we have infinite subsequences $\{\mu^{(k_i)}\}$ and $\{\phi_{(\mu^{(k_i)},\rho^{(k_i)})}(\boldsymbol{w}^{(k_i)})\}$ such that $\mu^{(k_i)} \leqslant \gamma_0\phi_{(\mu^{(k_i)},\rho^{(k_i)})}(\boldsymbol{w}^{(k_i)})$. Due to the fact that the limit of the subsequence $\{\phi_{(\mu^{(k_i)},\rho^{(k_i)})}(\boldsymbol{w}^{(k_i)})\}$ is zero, the limit of $\{\mu^{(k_i)}\}$ is also zero. We have $\lim_{k\to\infty} \mu^{(k)} = 0$, since $\{\mu^{(k)}\}$ is a bounded and non-increasing sequence. Therefore, both of the above cases lead to $\mu^* = 0$.

Compared with interior-point methods, IPRSOCP does not require iterative points to be in the interior of second-order cones. Taking advantage of this feature, we select the starting point as follows:

$$\boldsymbol{x}^{(0)} = A^\top(AA^\top)^{-1}\boldsymbol{b}, \tag{52a}$$

$$\boldsymbol{\lambda}^{(0)} = (AA^\top)^{-1}A\boldsymbol{c}, \tag{52b}$$

$$\boldsymbol{s}^{(0)} = \boldsymbol{c} - A^\top\boldsymbol{\lambda}^{(0)}. \tag{52c}$$

Equation (52) is actually the solution to two linear least squares problems.

It is easy to verify that this customized starting point always satisfies (31b) and (31c) during the iterative processes. In this way, we only need to consider the residuals of relaxed inequality constraints and

$$\psi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}) = \phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}),\ k = 0, 1, 2, \cdots. \tag{53}$$

Corollary 2 *Under Assumptions 1 and 2, if* $\{\rho^{(k)}\}$ *is upper bounded and the starting point is given in (52), then for* $\{\mu^{(k)}\}$ *and* $\{\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})\}$ *generated by Algorithm 1, we have*

$$\lim_{k\to\infty} \psi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)}) = 0, \quad \lim_{k\to\infty} \mu^{(k)} = 0.$$

Proof This conclusion can be obtained from (53) and Theorem 8.

6 Numerical Experience

To illustrate the effectiveness of Algorithm 1, we compare it with the open-source solvers SeDuMi [43], SDPT3 [45], and ECOS [13]. Algorithm 1 is implemented in MATLAB (Version R2017b), and the numerical tests are conducted on a server with the LINUX operating system (Ubuntu 20.04.1). The starting point $\left(\boldsymbol{x}^{(0)}, \boldsymbol{\lambda}^{(0)}, \boldsymbol{s}^{(0)}\right)$ is determined by (52). We initialize the penalty parameter $\rho^{(0)} = 2$. The barrier parameter $\mu^{(0)}$ is adaptively selected. In the Armijo line search method, we set $\tau = 0.05, \delta = 0.5$. The other parameters are set as follows: $\gamma_0 = 0.02,\ \sigma = 0.05,\ \eta = 10$.

The relative residuals of the primal and dual infeasibilities are defined as follows:

$$\text{pinf} = \frac{\|A\boldsymbol{x} - \boldsymbol{b}\|}{1 + \|\boldsymbol{b}\|}, \ \text{dinf} = \frac{\|\boldsymbol{c} - A^\top\boldsymbol{\lambda} - \boldsymbol{s}\|}{1 + \|\boldsymbol{c}\|}. \tag{54}$$

We claim to find an approximate optimal solution if the following condition is satisfied:

$$\max\{\text{pinf}, \text{dinf}, \mu\} \leqslant \epsilon. \tag{55}$$

In fact, since the iterative points of IPRSOCP satisfy (31b) and (31c), we only need to detect $\mu \leqslant \epsilon$. In general, we choose $\epsilon = 10^{-6}$. The maximum number of iterations is set to be 100.

As mentioned in [7, 44], the calculation of the primal direction $\Delta\boldsymbol{x}$ may not be very accurate due to the presence of rounding errors, i.e.,

$$\|A(\boldsymbol{x} + \Delta\boldsymbol{x}) - \boldsymbol{b}\| > \|A\boldsymbol{x} - \boldsymbol{b}\|$$

may occur. In this case, we make an orthogonal projection of $\Delta\boldsymbol{x}$ to the null space $\{\boldsymbol{u} : A\boldsymbol{u} = \boldsymbol{0}\}$, i.e., $\Delta\boldsymbol{x}$ is replaced by

$$(I - A^\top(AA^\top)^{-1}A)\Delta\boldsymbol{x}. \tag{56}$$

The Lasso and portfolio optimization problems from [39] are used to demonstrate the effectiveness and robustness of Algorithm 1.

6.1 Lasso

Firstly, we consider the ℓ_1-regularized least squares regression (Lasso) problem as follows:

$$\min \frac{1}{2}\|F\boldsymbol{x} - \boldsymbol{g}\|^2 + \mu\|\boldsymbol{x}\|_1, \tag{57}$$

where $F \in \mathbb{R}^{q\times p}$, $\boldsymbol{g} \in \mathbb{R}^q$, $\mu \in \mathbb{R}_+$ are the parameters, and $\boldsymbol{x} \in \mathbb{R}^p$ is the variable. The Lasso problem can be transformed into a SOCP problem [35]

$$\begin{aligned} \min\ & \frac{1}{2}w + \mu\mathbf{1}^\top\boldsymbol{t} \\ \text{s.t.}\ & \|(1-w; 2(F\boldsymbol{x}-\boldsymbol{g}))\| \leqslant 1+w, \\ & -\boldsymbol{t} \leqslant \boldsymbol{x} \leqslant \boldsymbol{t}, \end{aligned} \tag{58}$$

with variables $\boldsymbol{x} \in \mathbb{R}^p$, $\boldsymbol{t} \in \mathbb{R}^p$ and $w \in \mathbb{R}$.

We use the performance plot [12] as a tool to compare the efficiency of four solvers. The iterations and CPU time for solving Lasso problems are compared in Fig. 2. On more than 90% of problems, IPRSOCP takes fewer iterations than the other three solvers, and it takes the least CPU time to solve 50% of problems. Figure 2 further demonstrates that IPRSOCP and SDPT3 outperform SeDuMi and ECOS in terms of CPU time, with nearly identical performance on Lasso problems.

6.2 Portfolio Optimization

The portfolio problem [13, 39] is then considered, with the goal of achieving a trade-off in terms of risk and return:

$$\begin{aligned} \max\ & \boldsymbol{\mu}^\top\boldsymbol{x} - \gamma(\boldsymbol{x}^\top\Sigma\boldsymbol{x}) \\ \text{s.t.}\ & \mathbf{1}^\top\boldsymbol{x} = 1,\ \boldsymbol{x} \geqslant \mathbf{0}, \end{aligned} \tag{59}$$

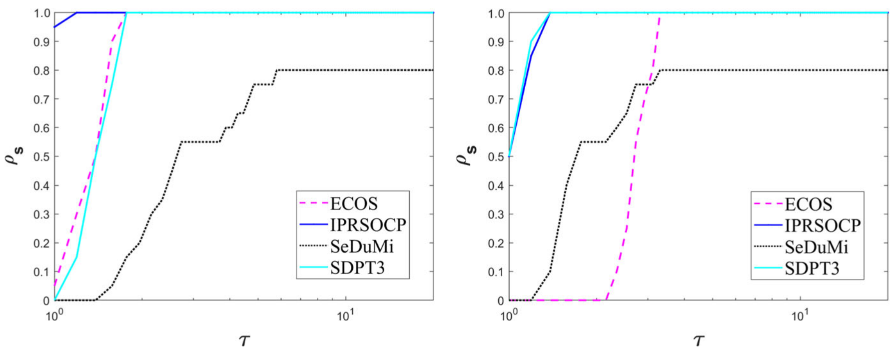

Fig. 2 Performance plots for Lasso problems with respect to iterations and CPU time.

where $\gamma > 0$ is the risk-aversion parameter, $\boldsymbol{\mu} \in \mathbb{R}^p$ and $\Sigma \in \mathbb{R}^{p \times p}$ are the expected value and covariance matrix of asset return, respectively. The variance matrix Σ is given by the factor model form,

$$\Sigma = FF^\top + D, \tag{60}$$

where $F \in \mathbb{R}^{p \times q}$ is the factor loading matrix and $D \in \mathbb{R}^{p \times p}$ is a diagonal matrix of asset-specific risk.

Problem (59) can be converted into a SOCP problem by using (60)

$$\begin{aligned} \min\ & -\boldsymbol{\mu}^\top \boldsymbol{x} + \gamma(t+s) \\ \text{s.t. } & \mathbf{1}^\top \boldsymbol{x} = 1,\ \boldsymbol{x} \geqslant \mathbf{0}, \\ & F^\top \boldsymbol{x} = \boldsymbol{u},\ \|(1-t; 2\boldsymbol{u})\| \leqslant 1+t, \\ & D^{\frac{1}{2}} \boldsymbol{x} = \boldsymbol{v},\ \|(1-s; 2\boldsymbol{v})\| \leqslant 1+s, \end{aligned} \tag{61}$$

with variables $\boldsymbol{u} \in \mathbb{R}^q$, $\boldsymbol{v} \in \mathbb{R}^p$, $t \in \mathbb{R}$, and $s \in \mathbb{R}$.

Figure 3 shows the performance plots of the four solvers for portfolio problems. IPRSOCP takes the fewest iterations on more than 80% of problems and takes the least CPU time on 70% of problems. It is worth noting that, for the portfolio problem, IPRSOCP and SeDuMi perform better than SDPT3 and ECOS in terms of CPU time. As a result, IPRSOCP is one of the best performing solvers for both Lasso and portfolio problems, demonstrating the robustness of IPRSOCP.

7 Conclusions

This paper develops IPRSOCP, a primal-dual interior-point relaxation algorithm that solves SOCP problems by utilizing a smoothing barrier augmented Lagrangian function. IPRSOCP does not require iterative points inside the interior region like interior-point methods, while it provides an explicit form of the Schur complement matrix that is not provided by existing non-interior-point methods. We find that the Schur complement matrix for IPRSOCP has a low-rank structure, which can be

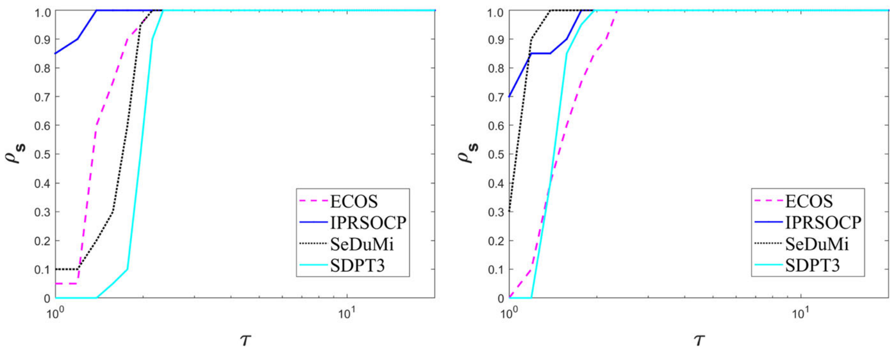

Fig. 3 Performance plots for portfolio problems with respect to iterations and CPU time.

exploited to enhance the stability and efficiency of matrix factorization. Under suitable assumptions, we prove the global convergence of the proposed algorithm. Numerical results show that IPRSOCP is comparable to the open-source solvers SeDuMi, SDPT3, and ECOS in terms of robustness and efficiency.

Appendix A: Some Proofs

Proof of Lemma 2

Proof Define the spectral decomposition $\boldsymbol{v}_i = \eta_1 \boldsymbol{d}_1 + \eta_2 \boldsymbol{d}_2$. Then

$$\begin{aligned}\boldsymbol{u}_i &= \left((\boldsymbol{s}_i - \rho \boldsymbol{x}_i)^2 + 4\rho\mu \boldsymbol{e}_i\right)^{\frac{1}{2}} \\ &= \left(\eta_1^2 \boldsymbol{d}_1 + \eta_2^2 \boldsymbol{d}_2 + 4\rho\mu \boldsymbol{d}_1 + 4\rho\mu \boldsymbol{d}_2\right)^{\frac{1}{2}} \\ &= \sqrt{\eta_1^2 + 4\rho\mu}\boldsymbol{d}_1 + \sqrt{\eta_2^2 + 4\rho\mu}\boldsymbol{d}_2.\end{aligned}$$

Therefore, we can obtain

$$\begin{aligned}\boldsymbol{z}_i &= \frac{\boldsymbol{u}_i - \boldsymbol{v}_i}{2\rho} = \frac{\sqrt{\eta_1^2 + 4\rho\mu} - \eta_1}{2\rho}\boldsymbol{d}_1 + \frac{\sqrt{\eta_2^2 + 4\rho\mu} - \eta_2}{2\rho}\boldsymbol{d}_2, \\ \boldsymbol{y}_i &= \frac{\boldsymbol{u}_i + \boldsymbol{v}_i}{2\rho} = \frac{\sqrt{\eta_1^2 + 4\rho\mu} + \eta_1}{2\rho}\boldsymbol{d}_1 + \frac{\sqrt{\eta_2^2 + 4\rho\mu} + \eta_2}{2\rho}\boldsymbol{d}_2.\end{aligned}$$

$\boldsymbol{z}_i$ and $\boldsymbol{y}_i$ share the same Jordan frame $\{\boldsymbol{d}_1, \boldsymbol{d}_2\}$ and are thus operator commutative.
Because the eigenvalues of $\boldsymbol{z}_i$ and $\boldsymbol{y}_i$ are non-negative, we have $\boldsymbol{z}_i,\ \boldsymbol{y}_i \in K_i$, and

$$\begin{aligned}&\boldsymbol{z}_i \circ \boldsymbol{y}_i \\ &= \left(\frac{\sqrt{\eta_1^2 + 4\rho\mu} - \eta_1}{2\rho}\boldsymbol{d}_1 + \frac{\sqrt{\eta_2^2 + 4\rho\mu} - \eta_2}{2\rho}\boldsymbol{d}_2\right) \circ \left(\frac{\sqrt{\eta_1^2 + 4\rho\mu} + \eta_1}{2\rho}\boldsymbol{d}_1 + \frac{\sqrt{\eta_2^2 + 4\rho\mu} + \eta_2}{2\rho}\boldsymbol{d}_2\right) \\ &= \frac{\mu}{\rho}\boldsymbol{d}_1 + \frac{\mu}{\rho}\boldsymbol{d}_2 = \frac{\mu}{\rho}\boldsymbol{e}_i,\ i = 1, 2, \cdots, n,\end{aligned}$$

which completes the proof.

Proof of Theorem 5

Proof The directional derivative of the merit function $\phi_{(\mu,\rho^{(k)})}(\boldsymbol{w})$ at the point $\left(\mu^{(k)}, \boldsymbol{w}^{(k)}\right)$ along the direction $\left(\Delta\mu^{(k)}, \Delta\boldsymbol{w}^{(k)}\right)$ is

$$\phi'_{(\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)}; \Delta\mu^{(k)}, \Delta\boldsymbol{w}^{(k)}\right) = -2\phi_{(\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)}\right). \tag{62}$$

The Taylor expansion of $\phi_{(\mu^{(k)}+\alpha\Delta\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)}+\alpha\Delta\boldsymbol{w}^{(k)}\right)$ with respect to α at $\alpha=0$ shows that

$$\begin{aligned}
&\phi_{(\mu^{(k)}+\alpha\Delta\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)}+\alpha\Delta\boldsymbol{w}^{(k)}\right)-\phi_{(\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)}\right)\\
&\quad=\alpha\phi'_{(\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)};\Delta\mu^{(k)},\Delta\boldsymbol{w}^{(k)}\right)+o(\alpha)\\
&\quad=-2\tau\alpha\phi_{(\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)}\right)-2(1-\tau)\alpha\phi_{(\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)}\right)+o(\alpha).
\end{aligned}\tag{63}$$

Since $\tau<1$ and $\phi_{(\mu^{(k)},\rho^{(k)})}\left(\boldsymbol{w}^{(k)}\right)>0$, then (45) holds for all sufficiently small $\alpha>0$.

Proof of Theorem 6

Proof By Lemma 8, we have $\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})\leqslant\phi_{(\mu^{(0)},\rho^{(0)})}(\boldsymbol{w}^{(0)})$ for all $k=1,2,\cdots$ By the definition of the merit function $\phi_{(\mu^{(k)},\rho^{(k)})}(\boldsymbol{w}^{(k)})$, we can obtain

$$\frac{1}{2}\|\boldsymbol{z}^{(k)}-\boldsymbol{x}^{(k)}\|^2\leqslant\phi_{(\mu^{(0)},\rho^{(0)})}(\boldsymbol{w}^{(0)}),$$

which together with Assumption 2 implies that $\{\boldsymbol{z}^{(k)}\}$ is bounded.

Define the spectral decomposition $\boldsymbol{v}_i=\eta_1\boldsymbol{d}_1+\eta_2\boldsymbol{d}_2$. By the definition of $\boldsymbol{y}_i$, we have

$$\boldsymbol{y}_i=\frac{\sqrt{\eta_1^2+4\rho\mu}+\eta_1}{2\rho}\boldsymbol{d}_1+\frac{\sqrt{\eta_2^2+4\rho\mu}+\eta_2}{2\rho}\boldsymbol{d}_2.$$

Since $\|\boldsymbol{d}_i\|=\frac{1}{\sqrt{2}}$, $i=1,2$, it is sufficient to prove that $\frac{\sqrt{\eta_1^2+4\rho\mu}+\eta_1}{2\rho}$ and $\frac{\sqrt{\eta_2^2+4\rho\mu}+\eta_2}{2\rho}$ are bounded. Note that

$$\begin{aligned}
&\frac{\sqrt{\eta_1^2+4\rho\mu}+\eta_1}{2\rho}\\
&\quad\leqslant\frac{|\eta_1|}{\rho}+\sqrt{\frac{\mu}{\rho}}\\
&\quad\leqslant\frac{|s_{i0}-\rho\boldsymbol{x}_{i0}|+\|\bar{\boldsymbol{s}}_i-\rho\bar{\boldsymbol{x}}_i\|}{\rho}+\sqrt{\frac{\mu}{\rho}}\\
&\quad\leqslant\frac{\sqrt{2}\|\boldsymbol{s}_i-\rho\boldsymbol{x}_i\|}{\rho}+\sqrt{\frac{\mu}{\rho}}\\
&\quad\leqslant\frac{\sqrt{2}\|\boldsymbol{s}_i\|}{\rho}+\sqrt{2}\|\boldsymbol{x}_i\|+\sqrt{\frac{\mu}{\rho}}
\end{aligned}$$

$$\leqslant \sqrt{2}\frac{\max\{1, \|\boldsymbol{x}_i\|\}}{\sigma} + \sqrt{2}\|\boldsymbol{x}_i\| + \sqrt{\frac{\mu}{\rho}}.$$

The third inequality holds because $|a| + |b| \leqslant \sqrt{2}\sqrt{a^2 + b^2}$. Thus, $\dfrac{\sqrt{\eta_1^2 + 4\rho\mu} + \eta_1}{2\rho}$ is bounded under Assumption 2. The boundedness of $\dfrac{\sqrt{\eta_2^2 + 4\rho\mu} + \eta_2}{2\rho}$ can be similarly deduced. So $\{\boldsymbol{y}^{(k)}\}$ is bounded.

Due to $\rho^{(k)}\boldsymbol{y}_i^{(k)}\boldsymbol{z}_i^{(k)} = \mu^{(k)}\boldsymbol{e}_i$, we have $\|\boldsymbol{y}_i^{(k)}\boldsymbol{z}_i^{(k)}\| = \dfrac{\mu^{(k)}}{\rho^{(k)}}$. Combined with the boundedness of $\{\boldsymbol{y}_i^{(k)}\}$ and $\{\boldsymbol{z}_i^{(k)}\}$, one has the desired inequalities.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflict of interest Yu-Hong Dai is an associate editor-in-chief for *Journal of the Operations Research Society of China* and he was not involved in the editorial review or the decision to publish this article. No potential Conflict of interest was reported by the authors.

References

[1] Alizadeh, F., Goldfarb, D.: Second-order cone programming. Math. Program. **95**(1), 3–51 (2003)
[2] Alizadeh, F., Haeberly, J.-P.A., Overton, M.L.: Primal-dual interior-point methods for semidefinite programming: convergence rates, stability and numerical results. SIAM J. Optim. **8**(3), 746–768 (1998)
[3] Ben-Tal, A., Nemirovski, A.: Robust convex optimization. Math. Oper. Res. **23**(4), 769–805 (1998)
[4] Boyd, S., Vandenberghe, L.: Convex Optimization. Cambridge University Press, New York (2004)
[5] Cai, Z., Toh, K.C.: Solving second-order cone programming via a reduced augmented system approach. SIAM J. Optim. **17**(3), 711–737 (2006)
[6] Chen, B., Harker, P.T.: A non-interior-point continuation method for linear complementarity problems. SIAM J. Matrix Anal. Appl. **14**(4), 1168–1190 (1993)
[7] Chen, X.D., Sun, D.F., Sun, J.: Complementarity functions and numerical experiments on some smoothing Newton methods for second-order cone complementarity problems. Comput. Optim. Appl. **25**(1), 39–56 (2003)
[8] Chi, C.Y., Li, W.C., Lin, C.H.: Convex Optimization for Signal Processing and Communications: from Fundamentals to Applications. CRC Press, Boca Raton (2017)
[9] Chi, X.N., Liu, S.Y.: A one-step smoothing Newton method for second-order cone programming. J. Comput. Appl. Math. **223**(1), 114–123 (2009)
[10] Dai, Y.H., Liu, X.W., Sun, J.: A primal-dual interior-point method capable of rapidly detecting infeasibility for nonlinear programs. J. Ind. Manag. Optim. **16**(2), 1009–1035 (2020)
[11] De Luca, T., Facchinei, F., Kanzow, C.: A theoretical and numerical comparison of some semismooth algorithms for complementarity problems. Comput. Optim. Appl. **16**, 173–205 (2000)
[12] Dolan, E.D., Moré, J.J.: Benchmarking optimization software with performance profiles. Math. Program. **91**(2), 201–213 (2002)
[13] Domahidi, A., Chu, E., Boyd, S.: ECOS: an SOCP solver for embedded systems. In: 2013 European Control Conference, pp. 3071–3076 (2013)

[14] El Ghaoui, L., Oustry, F., Lebret, H.: Robust solutions to uncertain semidefinite programs. SIAM J. Optim. **9**(1), 33–52 (1998)
[15] Engelke, S., Kanzow, C.: Predictor-corrector smoothing methods for linear programs with a more flexible update of the smoothing parameter. Comput. Optim. Appl. **23**(3), 299–320 (2002)
[16] Facchinei, F., Pang, J.S.: Finite-Dimensional Variational Inequalities and Complementarity Problems. Springer, New York (2003)
[17] Fukushima, M., Luo, Z.Q., Tseng, P.: Smoothing functions for second-order cone complementarity problems. SIAM J. Optim. **12**(2), 436–460 (2002)
[18] Gill, P.E., Kungurtsev, V., Robinson, D.P.: A shifted primal-dual penalty-barrier method for nonlinear optimization. SIAM J. Optim. **30**(2), 1067–1093 (2020)
[19] Goldfarb, D., Scheinberg, K.: Product-form Cholesky factorization in interior point methods for second-order cone programming. Math. Program. **103**(1), 153–179 (2005)
[20] Goldfarb, D., Polyak, R., Scheinberg, K., Yuzefovich, I.: A modified barrier-augmented Lagrangian method for constrained minimization. Comput. Optim. Appl. **14**, 55–74 (1999)
[21] Gondzio, J.: Warm start of the primal-dual method applied in the cutting-plane scheme. Math. Program. **83**(1–3), 125–143 (1998)
[22] Gondzio, J., González-Brevis, P.: A new warmstarting strategy for the primal-dual column generation method. Math. Program. **152**, 113–146 (2015)
[23] Hayashi, S., Yamashita, N., Fukushima, M.: A combined smoothing and regularization method for monotone second-order cone complementarity problems. SIAM J. Optim. **15**(2), 593–615 (2005)
[24] Helmberg, C., Rendl, F., Vanderbei, R.J., Wolkowicz, H.: An interior-point method for semidefinite programming. SIAM J. Optim. **6**(2), 342–361 (1996)
[25] John, E., Yıldırım, E.A.: Implementation of warm-start strategies in interior-point methods for linear programming in fixed dimension. Comput. Optim. Appl. **41**(2), 151–183 (2008)
[26] Kamath, A.G., Elango, P., Yu, Y., Mceowen, S., Carson III, J.M., Açıkmeşe, B.: Real-time sequential conic optimization for multi-phase rocket landing guidance. arXiv:2212.00375 (2022)
[27] Kanzow, C.: Some noninterior continuation methods for linear complementarity problems. SIAM J. Matrix Anal. Appl. **17**(4), 851–868 (1996)
[28] Kanzow, C., Ferenczi, I., Fukushima, M.: On the local convergence of semismooth Newton methods for linear and nonlinear second-order cone programs without strict complementarity. SIAM J. Optim. **20**(1), 297–320 (2009)
[29] Kojima, M., Shindoh, S., Hara, S.: Interior-point methods for the monotone semidefinite linear complementarity problem in symmetric matrices. SIAM J. Optim. **7**(1), 86–125 (1997)
[30] Kong, L.C., Sun, J., Xiu, N.H.: A regularized smoothing Newton method for symmetric cone complementarity problems. SIAM J. Optim. **19**(3), 1028–1047 (2008)
[31] Kuo, Y.J., Mittelmann, H.D.: Interior point methods for second-order cone programming and OR applications. Comput. Optim. Appl. **28**, 255–285 (2004)
[32] Liu, X.W., Dai, Y.H.: A globally convergent primal-dual interior-point relaxation method for nonlinear programs. Math. Comput. **89**(323), 1301–1329 (2019)
[33] Liu, X.W., Dai, Y.H., Huang, Y.K.: A primal-dual interior-point relaxation method with global and rapidly local convergence for nonlinear programs. Math. Methods Oper. Res. **96**(3), 351–382 (2022)
[34] Liu, X.W., Dai, Y.H., Huang, Y.K., Sun, J.: A novel augmented Lagrangian method of multipliers for optimization with general inequality constraints. Math. Comput. **92**(341), 1301–1330 (2023)
[35] Lobo, M.S., Vandenberghe, L., Boyd, S., Lebret, H.: Applications of second-order cone programming. Linear Algebra Appl. **284**(1–3), 193–228 (1998)
[36] Monteiro, R.D.: Primal-dual path-following algorithms for semidefinite programming. SIAM J. Optim. **7**(3), 663–678 (1997)
[37] Monteiro, R.D., Tsuchiya, T.: Polynomial convergence of primal-dual algorithms for the second-order cone program based on the MZ-family of directions. Math. Program. **88**(1), 61–83 (2000)
[38] Narushima, Y., Sagara, N., Ogasawara, H.: A smoothing Newton method with Fischer-Burmeister function for second-order cone complementarity problems. J. Optim. Theory Appl. **149**, 79–101 (2011)
[39] Odonoghue, B., Chu, E., Parikh, N., Boyd, S.: Conic optimization via operator splitting and homogeneous self-dual embedding. J. Optim. Theory Appl. **169**(3), 1042–1068 (2016)
[40] Qi, L.Q., Sun, D.F., Zhou, G.L.: A new look at smoothing Newton methods for nonlinear complementarity problems and box constrained variational inequalities. Math. Program. **87**(1), 1–35 (2000)

[41] Skajaa, A., Ye, Y.Y.: A homogeneous interior-point algorithm for nonsymmetric convex conic optimization. Math. Program. **150**, 391–422 (2015)
[42] Smale, S.: Algorithms for solving equations. Proceedings of the International Congress of Mathematicians, Berkeley (1986)
[43] Sturm, J.F.: Using SeDuMi 1.02, a Matlab toolbox for optimization over symmetric cones. Optim. Methods Softw. **11–12**, 625–653 (1999)
[44] Todd, M.J., Toh, K.C., Tütüncü, R.H.: On the Nesterov-Todd direction in semidefinite programming. SIAM J. Optim. **8**(3), 769–796 (1998)
[45] Toh, K.C., Todd, M.J., Tütüncü, R.H.: On the implementation and usage of SDPT3-a Matlab software package for semidefinite-quadratic-linear programming, version 4.0. In: Handbook on Semidefinite. Conic and Polynomial Optimization, pp. 715–754. Springer, Boston (2012)
[46] Tsuchiya, T.: A convergence analysis of the scaling-invariant primal-dual path-following algorithms for second-order cone programming. Optim. Methods Softw. **11**(1–4), 141–182 (1999)
[47] Wright, S.J.: Primal-dual Interior-point Methods. SIAM, Philadelphia (1997)
[48] Xu, S., Burke, J.V.: A polynomial time interior-point path-following algorithm for LCP based on Chen-Harker-Kanzow smoothing techniques. Math., Program. **86**(1) (1999)
[49] Yildirim, E.A., Wright, S.J.: Warm-start strategies in interior-point methods for linear programming. SIAM J. Optim. **12**(3), 782–810 (2002)
[50] Zhang, R.J., Liu, X.W., Dai, Y.H.: IPRSDP: a primal-dual interior-point relaxation algorithm for semidefinite programming. Tech. Rep. (2022)
[51] Zhang, R.J., Liu, X.W., Dai, Y.H.: IPRQP: a primal-dual interior-point relaxation algorithm for convex quadratic programming. J. Global Optim. **87**(2), 1027–1053 (2023)